NATIONAL UNIVERSITY OF LESOTHO

Department of Physics and Electronics



Laboratory manual

B.Eng and B.Sc in Electronics
Academic year: 2009/2010

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Student no: .....

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LABORATORY EXECUTION GUIDE:

A. READ UP IN ADVANCE

Prior preparation for an experiment is the basis for good experiment execution. You will be expelled from the laboratory if you are adjudged not to be properly/fully prepared for the experiment

B. CHOICE OF APPARATUS

You should select apparatus, which will measure the required results over a suitable **range** of values to a sufficient but not excessive accuracy. In particular, you should select ammeters and voltmeters of the correct impedance so that introducing them into the circuit does not change the current or voltage to be measured. In most cases, appropriate equipment will be already set up for the respective experiments. ***IF IN DOUBT ASK!***

C. EXPERIMENTAL PROCEDURE

You will be required to keep a proper and tidy laboratory log book which means ***all your laboratory work will be kept in the RED laboratory books*** (not scrap papers). N.B. "An experimental log-book is the tool you use to log your data as you perform the experiment". In other words data capturing and all events relevant to the experiment are recorded in the experiment book as and when you perform the experiment. After completion of your lab work, the demonstrator will sign the page on which the work was done (as a means of authenticating your experimental work/results).

<u>WARNING:</u> - NEVER, EVER USE TIPP-EX, TO ERASE YOUR LABORATORY ENTRIES! THE PENALTY WILL BE <u>ZERO</u> MARKS IF YOU DO!

D. GUIDELINES TO LABORATORY REPORT WRITING

All laboratory WORK will be written in the *"laboratory report books" ONLY.* A standard lab report will include the following:

Title: This should indicate the subject of the report as *clearly and briefly* as possible and the *date* of the report.

Abstract: This should give a clear and concise summary of the purpose and contents of the report, and the most significant results and conclusions. *Five to ten lines will suffice*.

Introduction: The principal objective of the introduction is to acquaint the reader with the problem and to point out the purpose and significance of the report. The

introduction should be logically developed to explain the nature of the problem and to provide any background information and typical applications needed by the reader.

Theory: This is an important section that marks the beginning of the main body of the report. It should contain all the relevant theory and any other detailed technical information needed by the reader to understand the rest of the report. Where diagrams and equations are used to assist in the development of the theory, these should be properly introduced in the text - never leave the reader having to guess why a diagram or equation has been put in a report.

Experimental results: This section should show the results of the experiments. Any modifications done to the experimental set-up (e.g. usage of different component values from the ones stipulated in lab manual) should be presented here, together with their associated effect on the theoretical predictions.

Discussion/Interpretation of results: This section should analyze the results, comparing the theoretical predictions with the measured experimental results. Sources of possible errors should be explained. It is important that you actually discuss and assess your results, and do not simply present a re-hashed version of earlier theoretical predictions. You must clearly set out your interpretation of the results to show how they confirm (or fail to confirm) the aims of the practical, e.g. try to explain why a curve appears as it does. Do not merely say 'it is linear', but 'it is linear because …'. For example, extrapolate graphs to zero and make inferences from the result, or if an instrument is suspect, calculate that effect on the result.

* If a result is believed to be inaccurate due to an instrument reading, make an assessment of the experimental error and calculate the effect on the result in question.

* Do not say "results are not as expected" unless you clearly told the reader what was expected and in what way they differ.

* Do not say "results are not very accurate" or "results are inaccurate" unless you can back up your statement by giving an estimate of accuracy. The same goes for words like "poor", good", etc

* Do not present a verbal version of your graphs (i.e. merely describing their shape), say why they appear as they do.

Conclusion: The conclusion should state concisely what is shown by the work and significance of the findings. Avoid exaggeration and remain focused on what can genuinely be concluded from your results. Conclusions are NOT:

- Results “were as expected” or “not as expected”. Even if you have stated your expectations (and students rarely do), your reader does not want to go and look them up in order to find out what conclusions you are making.
- That you learned a lot, that the practical was interesting, that the apparatus broke down, that the practical was unsuccessful, etc – these were not the objectives of the experiment, or
- That the results were good, bad, inaccurate, etc, unless you have argued these points in the discussion and you have quoted actual figures in the conclusion.

References: Referencing is a particularly important aspect of technical report writing, and there are some very serious implications if it is not carried out correctly. Referencing serves two different important functions, (i) to point the reader to sources of further information; (ii) to acknowledge all other sources of information used by the author in compiling his or her own report

E. GENERAL

Units & quantities: Metric (SI) units should be used. The omission of physical units in an engineering/scientific report is unacceptable.

Diagrams, tables and graphs: Diagrammatic material should be presented in a format that permits it to be read either with the report the usual way up, or with the report turned clockwise through 90˚. All figures should be numbered with the text below/above the figure e.g. “Figure 1 Armature voltage as a function of field current”. Tables should be numbered with the text above the table e.g. “Table 1 Armature voltage as a function of field current”.

Equations: All equations must be numbered sequentially as shown below in round brackets

“The output voltage of the amplifier is given by

$$V_o = A_v(V_1 + V_2) \qquad (1)$$

where A_v *= the voltage gain*
V_1 *= input voltage 1*
V_2 *= input voltage 2*

Plotting the graphs: Each axis must be labelled with the name of the variable and its units e.g. Time (s) and Output Voltage (V). The scales of the axes should be appropriate for a well-proportioned graph filling a reasonable portion of a page. Axes should be marked in intervals corresponding to 1, 2.5 or 10 units (or these times powers of 10). Other interval sizes make it extremely difficult to read between divisions on decimal graph paper.

Grammatical style: It is normal to write reports in the third person, passive voice. Use of the first person (I, we, me, us, etc) may be acceptable, but usually only in exceptional cases. You should use the past tense whenever you are reporting on things that were done, or decisions that were made during an assignment, design project or practical. You should use the present tense when describing theory and any other issues that are always true in the present.

A. Electronic Measurements & Instrumentation – EE3300

EMI 1 – Introduction to Electronic Measurements

OBJECTIVES

1. To understand the basic operation of the function generator, oscilloscope and multimeter.

2. To use the oscilloscope to observe signals created by various sources.

EQUIPMENT

Function generator; Oscilloscope; Multimeter;
Step-down transformer; Microphone;
4 Coaxial cables with BNC connectors;
1 T connector;
1 BNC-banana adapter

PROCEDURE

1. Familiarize yourself with the main controls on the function generator: the power switch, the function frequency selector, the function frequency multiplier, the offset knob, the amplitude knob, and the output function shape. Using the controls, set the generator to create a sine wave with a frequency of 3 kHz.

2. Next, familiarize yourself with the basic settings on the oscilloscope: the power button, the vertical scale knob, and the horizontal scale knob. Ignore the other buttons for now.

3. Connect one of the oscilloscope probes to the output wire for the function generator. Turn both instruments on. Adjust the vertical and horizontal knobs on the scope to get a clear view of the signal. *Sketch the diagram*.

4. Adjust the knobs on the function generator to see what different signals (square, triangular) look like. *Also sketch the diagrams*.

(a) Specifically, what are the minimum and maximum signal amplitudes that can be created by your function generator?

(b) What are the minimum and maximum signal frequencies it can produce?

5. Now that you have a basic understanding of how to work the oscilloscope, we can do something interesting. Let's look at the electrical signal that comes from a standard wall socket. To prevent you from being electrocuted, we will use a transformer to step down the voltage. You can think of this as a box which reduces

the voltage (in this case by a factor of 10). Connect the scope probe to the transformer output and then plug the transformer into the wall.

(a) *Sketch the signal*.

(b) What is its amplitude and frequency?

6. Let's look at another type of signal on the oscilloscope: sound waves. Unplug the scope probe from the function generator, and plug it instead into the microphone on your bench. The microphone converts sound waves in the air into electrical signals that can be analyzed on the scope.

7. Sing a note into the microphone and adjust the scope so that you can clearly see the electrical signal. Now, talk into the microphone and see what the signal looks like. Say a particular vowel, let's say the `e' in hello into the microphone.

(a) *Sketch the resulting signal and use the scope to figure out the frequency*.

(b) Compare the frequency of your voice to that of others in the lab. What is the range of frequencies you find?

(c) Scream a high pitch into the microphone. *Sketch the shape of the signal produced*. How does it differ from the low pitch signal?

8. Using a BNC-T connector, connect the output of the function generator (FG) to the multimeter while leaving the oscilloscope connected. For the multimeter connection, use a coaxial cable with a BNC plug on one end and a BNC-to-banana plug adapter on the other end. The positive lead of the banana plug adapter should be in the top right jack of the multimeter and the negative in the middle right jack. In order to tell which side of the banana plug is negative, look for a small ridge on one side. This side is the ground side. On some plugs, the letters GND are even written on this ridge. This ridge is called an ear, leading to the “ear to the ground” rule. Among other things, the multimeter is useful for measuring **root-mean-square** (RMS) voltage. The RMS value is often defined as:

$$V_{rms} = 0.707\, V_p \qquad (1)$$

where V_p is the peak voltage

Turn on the multimeter and measure the RMS voltage. Change the amplitude of the FG to 5 V_{pp} (peak-to-peak) and measure the corresponding RMS voltage.

9. Now investigate an AC signal with a DC offset. Change the FG back to the 2-V sine wave. Press the *Sine Wave* button on the top row and hold it. The FG is now set to output only a DC voltage. Adjust the output voltage to 1 V by turning the dial. The oscilloscope should now display a DC voltage at +1 V. Press the sine wave key. The

FG is now producing a 1 kHz sine wave with a 1 V DC offset. Check to make sure the o-scope coupling is DC. Notice the ground indicator to the right of the oscilloscope screen showing the 0 V level. Measure the RMS voltage using both the multimeter and the oscilloscope. Change the channel 1 coupling to AC and observe what happens to the oscilloscope's RMS reading. What does the AC coupling do? Repeat the RMS measurements for a 10 V_{pp} sine wave with and without a 5 V DC offset. Remove the offset by reversing the procedure for changing the DC offset.

10. Change the output waveform to a square wave (10 V_{pp}, no DC offset) by pressing the square wave key in the top row. Measure the RMS voltage with the multimeter and the peak-to-peak voltage with the oscilloscope. (Make sure % duty = 50 on the FG.)

11. Repeat the RMS voltage measurement with a triangle wave (10 V_{pp}, no DC offset). Again, record all readings.

DISCUSSION QUESTIONS

1. Explain the purpose of the following oscilloscope feature: AC, DC, GND coupling of the signal.

2. Describe what rms voltage means. For a sine wave, show that V_{rms} is equal to the following (work out the integral in your lab notebook):

$$V_{rms} = \left[\frac{1}{2\sqrt{2}}\right] V_{pp} \qquad (2)$$

3. When there is a DC offset in a signal, why does the multimeter give a different value for the rms voltage than the scope?

4. Explain the purpose of the following oscilloscope features: Auto level, Auto, Normal and Single trigger modes, AC/DC trigger coupling.

5. What scope settings (trigger, V/div, and time/div) would you use to capture a transient (single shot) signal consisting of two 1 Volt pulses, 100 ns in duration and separated by 10 ms? First sketch out what you'd like the scope screen to look like, then determine the settings to achieve this screen display.

EMI 2 – Passive Filters

OBJECTIVES

1. Plot the gain-frequency response of a first-order RC low-pass filter, RC high-pass filter, and LC series resonant band-pass filter.
2. Determine the cut-off frequency and roll-off of an RC low-pass filter and RC high-pass filter, the centre frequency and the bandwidth of the LC band-pass filter.
3. Determine how the value of R and C affects the cut-off frequency of an RC low-pass and high-pass filters.

EQUIPMENT

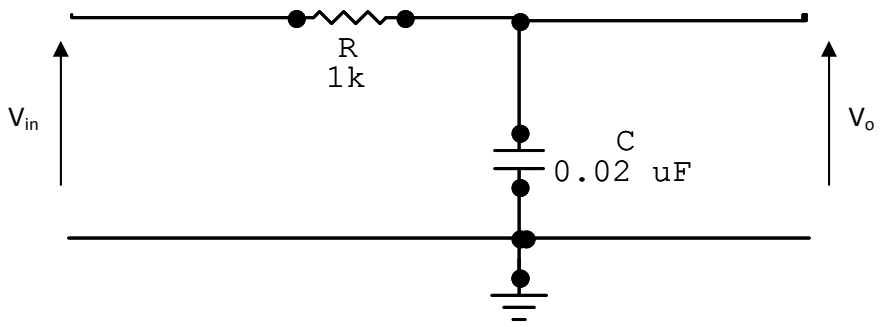
One function generator;
One dual-trace oscilloscope;
Breadboard;
Capacitors: 0.02 µF, 0.04 µF, 0.1 µF, 0.25 µF
Resistors: 100 Ω, 200 Ω, 1 kΩ, 2 kΩ, 4 kΩ, 5 kΩ, 200 kΩ
Inductors: 50 mH, 100 mH

PROCEDURE

[You will need to plot the frequency response curves by making measurements at different frequencies using an oscilloscope and plot the curves on semilog graph paper.]

Part 1

Figure 1 - Low-pass RC filter



1. Set up the circuit of Figure 1 and measure the voltage gain (V_o/V_{in}) for frequencies between 100 Hz and 1 MHz. Sketch the curve of frequency vs. voltage gain in dB.

2. Mark a point on the curve that is 3 dB down from the dB gain at 100 Hz. Record the frequency (cut-off frequency, f_c) on the curve plot.

3. Calculate the expected cut-off frequency based on the circuit component values in Figure 1.

4. Mark a point on the curve that is as close as possible to ten times f_c. Record the dB gain and frequency (f_2) on the curve plot. What is the roll-off in dB per decade?

Part 2

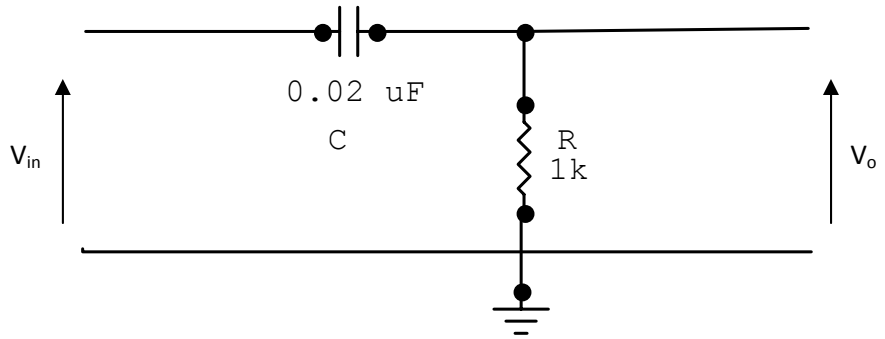
5. Set up the circuit of Figure 2 and measure the voltage gain for frequencies between 100 Hz and 1 MHz. Sketch the curve of frequency vs. voltage in dB.

6. Mark a point on the curve that is 3 dB down from the dB gain at 1 MHz. Record the frequency (cut-off frequency, f_c) on the curve plot.

7. Calculate the expected cut-off frequency based on the circuit component values in Fig. 2.

8. Mark a point on the curve that is as close as possible to one-tenth f_c. Record the dB gain and frequency (f_2) on the curve plot. What is the roll-off in dB per decade?

Figure 2 - High-pass RC filter



Part 3

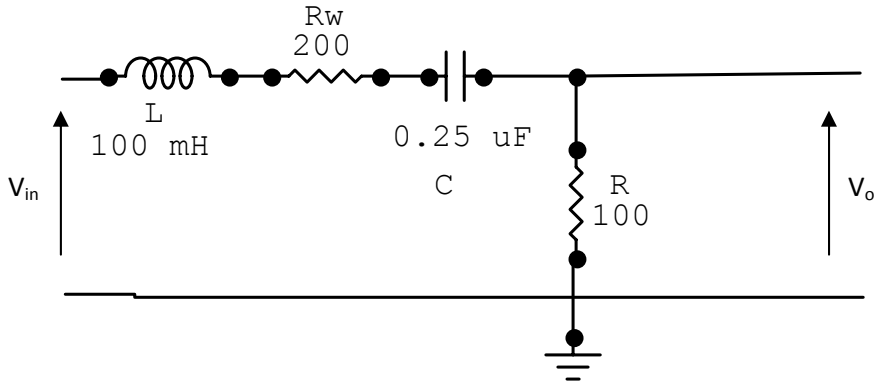
9. Set up the circuit of Figure 3 and measure the voltage gain in dB for frequencies between 500 Hz and 2 kHz. Sketch the curve plot.

10. Mark a point on the centre of the curve at its peak point. Record the centre frequency (f_o) and voltage gain in dB on the curve plot.

11. Mark a point on the left side of the curve that is 3 dB down from the dB gain measured at frequency f_o. Record the approximate frequency (low cut-off frequency, f_{c1}) on the curve plot. Next, mark a point on the right side of the curve that is 3 dB down from the dB gain measured at frequency f_o. Record the approximate frequency (high cut-off frequency, f_{c2}) on the curve plot.

12. Based on the values of f_{c1} and f_{c2} measured on the curve plot, calculate the centre frequency (f_o), the quality factor (Q_s) and the bandwidth of the series resonant band-pass filter.

Figure 3 - LC Series Resonant Band-pass Filter



13. Based on the circuit component values in Figure 3, calculate the expected centre frequency (f_o) of the series resonant band-pass filter.

14. Based on the circuit quality factor (Q_s) and the centre frequency (f_o), calculate the expected bandwidth (BW) of the series resonant band-pass filter.

EMI 3 – Active Filters

OBJECTIVES

1. Plot the gain-frequency response and determine the cut-off frequency of second order (two-pole) low-pass and high-pass active filters.
2. Plot the gain-frequency response curve and determine the centre frequency for active band-pass and band-stop filters.
3. Determine the quality factor (Q) and bandwidth of active band-pass and band-stop filters.
4. Determine the roll-off in dB per decade for a second-order (two-pole) filter.
5. Plot the phase-frequency response of a second-order filter.

EQUIPMENT

One function generator;
One dual-trace oscilloscope;
One LM741 op-amps;
Breadboard;
Capacitors: two 0.001 µF, one 1 pF, two 0.01 µF, two 0.05 µF, one 0.1 µF
Resistors: one 1 kΩ, one 5.86 kΩ, two 10 kΩ, one 13 kΩ, one 27 kΩ, two 30 kΩ, two 54 kΩ, one 100 kΩ

PROCEDURE

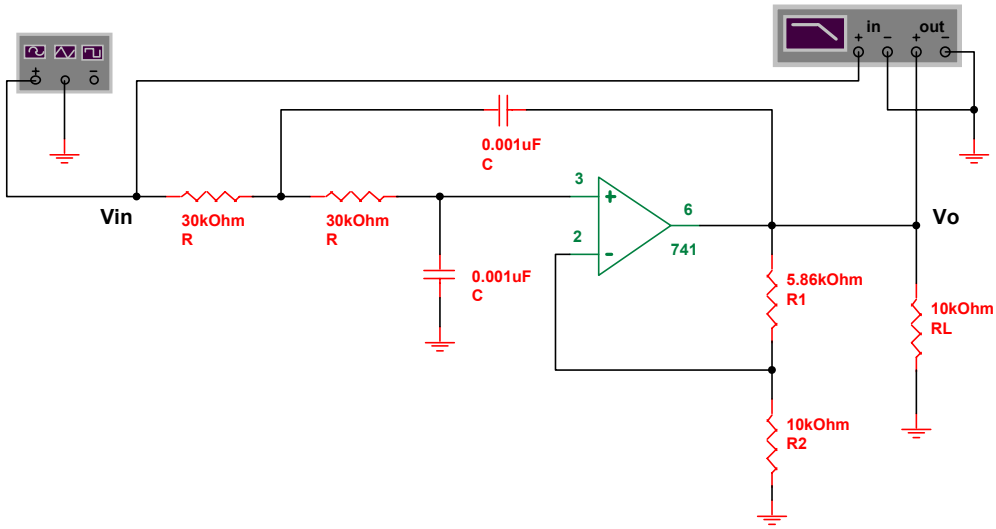
[*You will need to plot the frequency response curves by making dB gain measurements at different frequencies between 100 Hz and 100 kHz and plotting the curves on semilog graph paper*.]

The two-pole Sallen-Key Butterworth filters require a passband voltage gain of 1.586 to produce the Butterworth response. Therefore,

$$A_v = 1 + \frac{R_1}{R_2} = 1.586 \qquad \textit{and} \qquad \frac{R_1}{R_2} = 0.586$$

Part 1

Figure 1 - Second-order Sallen-Key Low-pass Butterworth filter



1. Set up the circuit of Figure 1 and measure the voltage gain in dB for frequencies between 100 Hz and 100 kHz. Sketch the curve plot.

2. Calculate the expected voltage gain (A) on the flat part of the curve for the low-pass Butterworth filter based on the circuit component values in Figure 1.

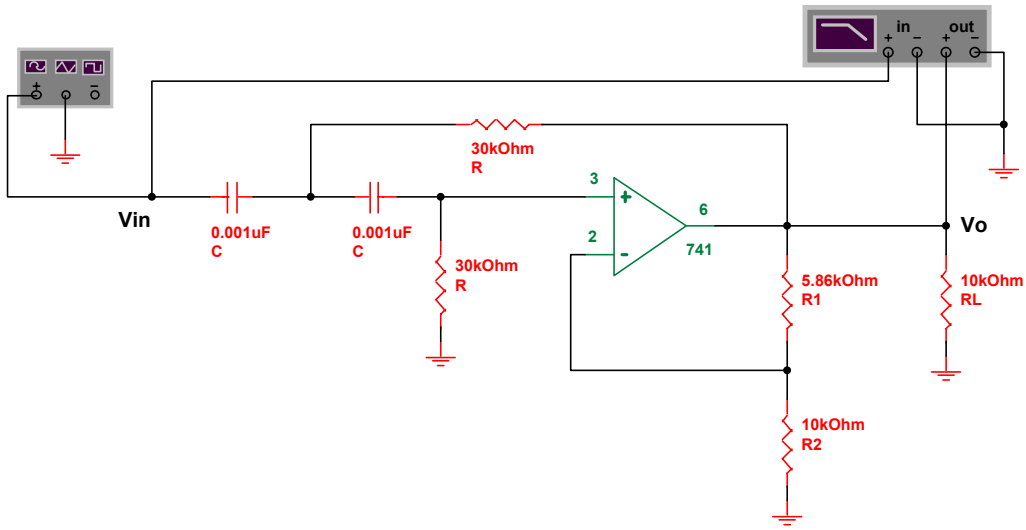
3. Mark a point on the curve that is 3 dB down from the dB gain at low frequencies. Record the dB gain and the frequency (cut-off frequency, fc) on the curve plot.

4. Calculate the expected cut-off frequency based on the circuit component values.

5. Mark a point on the curve that is as close as possible to ten times fc. Record the dB gain and frequency (f_2) on the curve plot.

Part 2

Figure 2 - Second-order Sallen-Key High-pass Butterworth filter



6. Set up the circuit of Figure 2 and measure the voltage gain in dB for frequencies between 100 Hz and 100 kHz. Sketch the curve plot.

7. Calculate the expected voltage gain (A) on the flat part of the curve for the high-pass Butterworth filter based on the circuit component values in Figure 2.

8. Mark a point on the curve that is 3 dB down from the dB gain at high frequencies. Record the dB gain and the frequency (cut-off frequency, f_c) on the curve plot.

9. Calculate the expected cut-off frequency based on the circuit component values.

10. Mark a point on the curve that is as close as possible to one-tenth f_c. Record the dB gain and frequency (f_2) on the curve plot.

11. Take measurements up to 50 MHz and draw the resulting curve. Measure the upper cut-off frequency and record the value on the curve plot. Why is this the case?

B. Analogue Circuit Design – EE3301

ACD 1 – Common Emitter Amplifier

OBJECTIVE

To investigate the operation of a common-emitter NPN transistor amplifier.

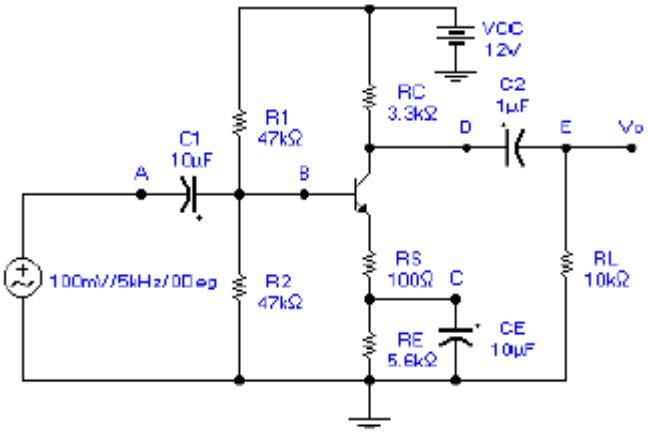
Some Circuit Notes:

1) The transistor is a general-purpose NPN transistor (2N3904, 2N2222), or equivalent.

2) Assume Beta = 100, V_{BE} = 0.7V, r'e = 25mv/I_E in the prelab analysis.

3) The computer simulations below refer to the use of *Electronics Workbench*.

Figure 1 - NPN Common Emitter Amplifier



Part 1: Prelab Analysis

For the circuit shown, predict the following DC parameters: I_E, V_E, V_B, V_C, and V_{CE}, using the "approximate method". Draw the ac equivalent circuit, and predict the voltage gain (A_v). Also draw the "composite" waveforms expected at points A, B, C, D, and E in the circuit, assuming V_{in} (at point A) is a 100 mV peak, 5 kHz sine wave.

Part 2: Computer simulation using Electronics Workbench:

Draw the circuit shown (Figure 1) using Electronics Workbench. Connect the amplifier input to a sinusoidal signal generator as shown in the figure. Use the DVM (digital voltmeter) in Electronics Workbench to measure the DC voltages on the transistor terminals, and compare these to the predicted and measured values. Then

connect the oscilloscope to the amplifier's input and output terminals and compare these "measured" results to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal; for example, by temporarily removing C_E , you can see that the output signal dramatically decreases; by "shorting out" the emitter swamping resistor Rs, the output signal (and non-linear distortion) increase significantly. By changing the base bias resistors, it may be possible to further optimize the amplifier for a larger peak output signal.

Part 3: Lab Measurements

Build the circuit shown and verify ALL of the dc and ac predictions above. Use DC coupling and dual-trace on the oscilloscope as appropriate. Then temporarily remove C_E, and remeasure the ac output voltage. Compare your results to your predictions.

EQUIPMENT

Digital Multimeter;
Function Generator;
Dual-trace scope;
Breadboard with DC power supply;
2N3904 or 2N2222 bipolar transistor;
Capacitors (2 x 10 µF and 1 µF);
Resistors (100 Ω, 3.3 kΩ, 5.6 kΩ, 10 kΩ, 2 x 47 kΩ)

Comments:

Compare your experimental and computer simulation results with your prelab calculations. Explain any significant discrepancies. Also, explain the purpose or function of the various components in the circuit; for example, what is the purpose of the 100 Ω resistor at the emitter? the emitter bypass capacitor? etc…

ACD 2 – Op-Amps and the Sound Amplifier

OBJECTIVES

1. To understand how op-amps amps can be used to amplify an electrical signal;
2. To use op-amp circuits in building a basic sound amplifier.

EQUIPMENT

Breadboard with DC Power supply;
LM741 Op-amp;
Resistors – 1k, 10k, 100k;
Transformer (step-down);
Audio speaker

PROCEDURE

1. The power supplies at your workbench only go between ground and some positive voltage, V. Thus, they are considered *one-sided*. Op-amps, however, require a *two-sided* power supply one that goes between -V and +V for some voltage V. To get a two-sided voltage, you will setup a voltage divider as drawn in the following schematic. This will split the output of the power supply into two voltages, with our ground as the point between the two.

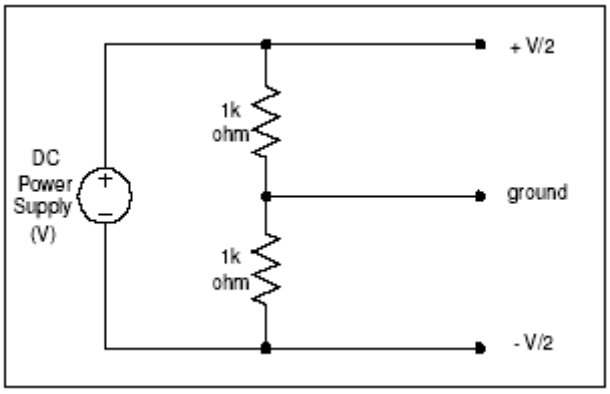


Figure 1: Voltage divider circuit.

2. The op-amp is in an integrated circuit chip which can be placed directly on your breadboard. There are many different types of op-amps, but the one we will be using is called the LM741 General Purpose Op-Amp, or '741 for short. Look at the '741 data sheet to see the uses for the various pins. Wire an op-amp circuit in the *inverting configuration* using a 100 k- resistor and a 10 k- resistor. Plug the function generator into the input of the circuit, set the frequency to 1 kHz and the amplitude to the

default (100 mV peak-to-peak). Attach one probe of the oscilloscope to the op-amp input and one probe to the op-amp output.

(a) Sketch the input and output signals on the same plot.

(b) What is the gain of the amplifier?

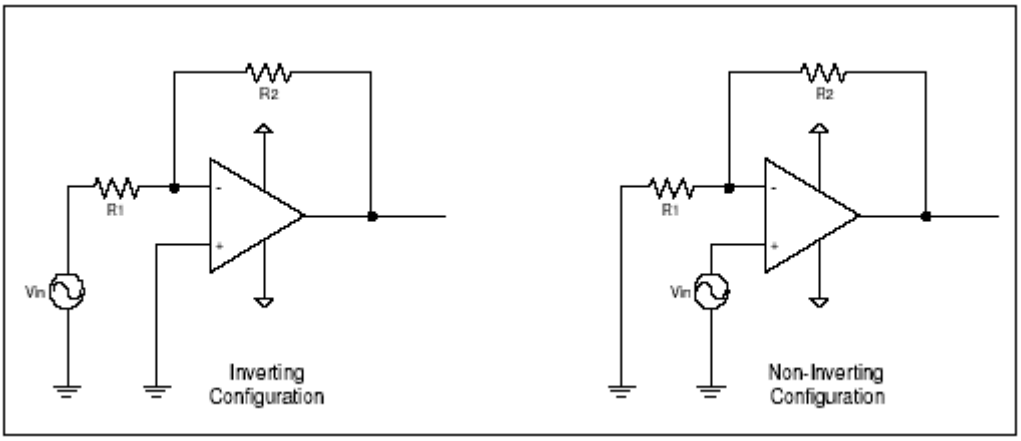


Figure 2: Review of op-amp amplifier configurations.

3. Now, increase the amplitude to its maximum value.

(a) What happens to the shape of the output?

(b) What is the maximum voltage that can be output by the op-amp?

4. Change the circuit used above to the *non-inverting configuration*.

(a) Sketch the input and output signals on the same plot.

(b) What is the gain of the amplifier now?

5. Now let's amplify something interesting- sound. First, plug the microphone outputs directly into the speaker inputs. Speak into the microphone and listen carefully to the speaker (put it against your ear). Do you hear anything?

6. It is clear that the signals created by the microphone have to be amplified before they get sent to the speaker if we are to get any sound out of them. Starting with the op-amp circuit used in step 4 above, replace one of the resistors to get a gain of around 100.

7. Now, plug the microphone into the op-amp input, one of the scope probes into the op-amp input, and the other scope probe into the op-amp circuit output. Talk or sing into the microphone, and observe the scope output.

(a) How well does the circuit work in amplifying voice signals?

8. Now, hook the speaker inputs to the op-amp outputs. Speak into the microphone, listen to the speaker output, and look at the output on the scope.

(a) Does the output sound realistic?

(b) Sketch the op-amp input and output signals.

9. The problem encountered above is that the speaker we are using has a very low resistance (R = 8 Ω). By Ohm's Law then, our poor op-amp has to output a lot of current to maintain the high output voltages it is producing, something it doesn't like to do. Thus, the signal gets distorted, something that is not supposed to happen in a typical audio amplifier. This is the equivalent of trying to start a car with an AA battery- it doesn't work too well!

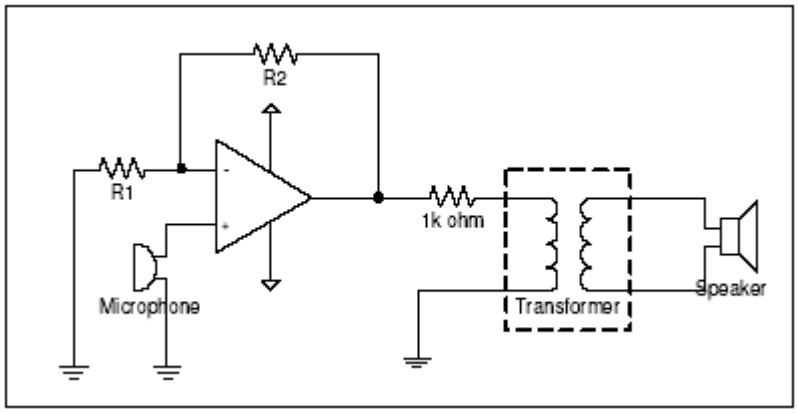


Figure 3: Schematic of circuit with transformer.

10. To help resolve this problem, there are several things one can do. One solution would be to add a second *stage* to the amplifier, one that doesn't have a large gain, but can output a lot of power. Multistage amplifiers are common in electrical engineering- the op-amp itself is actually a multistage amplifier made out of transistors. In this case, however, we can do something even easier- add a transformer. A transformer is a device which can step-up or step-down the voltage of an AC signal. Another feature of a transformer is that it can adjust the load resistance attached to a particular device. In this case, by putting in a transformer between the op-amp output and the speaker, the op-amp will act as though it has a 1000 - resistor attached to it, reducing the distortion problems seen previously.

11. Connect the transformer between the op-amp output and the speaker.

As shown in the figure above, the transformer should also have a resistor connected on the op-amp side (so that the op-amp load is high enough even when there is no signal). Be sure to connect the transformer correctly- unlike a resistor, it is not a symmetric device (the speaker and op-amp sides are different).

(a) How does the sound output compare to that of the amplifier without a transformer?

(b) Sketch the input and output waveforms. Is the new amplifier configuration free of distortion?

ACD 3 – Oscillators (Carrier Generators)

OBJECTIVES

1. Measure the frequency of oscillation and learn what determines the frequency of oscillation for an oscillator.
2. Learn how the LC filter capacitance ratio affects the amplifier voltage gain and the oscillator peak-to-peak output voltage in a Colpitts oscillator.

EQUIPMENT

DC power supply
2N3904 bipolar transistor
Signal Generator
Dual-trace oscilloscope
Phase Shift Oscillator Chassis
Capacitors: 0.001 µF, 0.01 µF, 3 x 0.1 µF, 500 pF
Inductors: 5 µH, 10 µH
Resistors: 3 x 2K, 15 K, 200 K

THEORY

An oscillator is a circuit that converts dc power supply voltage into an ac output voltage. A popular LC oscillator is the Colpitts oscillator, shown in Fig. 2. In order for oscillation to be sustained, the total voltage gain around the closed loop (AB) must equal unity (1) at the oscillation frequency. Therefore,

$$AB = 1$$

Where B is the oscillator feedback ration and A is the amplifier voltage gain.

Because the oscillator feedback ratio (B) is C_1/C_2 at the resonant frequency, the amplifier voltage gain (A) needed to sustain oscillation can be calculated from

$$AB = 1 \quad \Rightarrow \quad A\left(\frac{C_1}{C_2}\right) = 1 \quad \Rightarrow \quad A = \frac{C_2}{C_1}$$

The resonant frequency (f) of the LC filter determines the frequency of oscillation and is approximated from

$$f = \frac{1}{2\pi\sqrt{LC_T}}$$

where $C_T = C_1C_2 / (C_1 + C_2)$.

The voltage gain of the amplifier (A) is equal to the ac peak-to-peak output voltage (V_o) divided by the ac peak-to-peak input voltage (V_{in}). Therefore,

$$A = \frac{V_o}{V_{in}}$$

PROCEDURE

Part 1 – Phase shift oscillator

1. The phase shift oscillator chassis consists of a stabilised transistor amplifier and a feedback loop consisting of a potential divider P and a phase network. The feedback loop can be disconnected by means of a switch S.

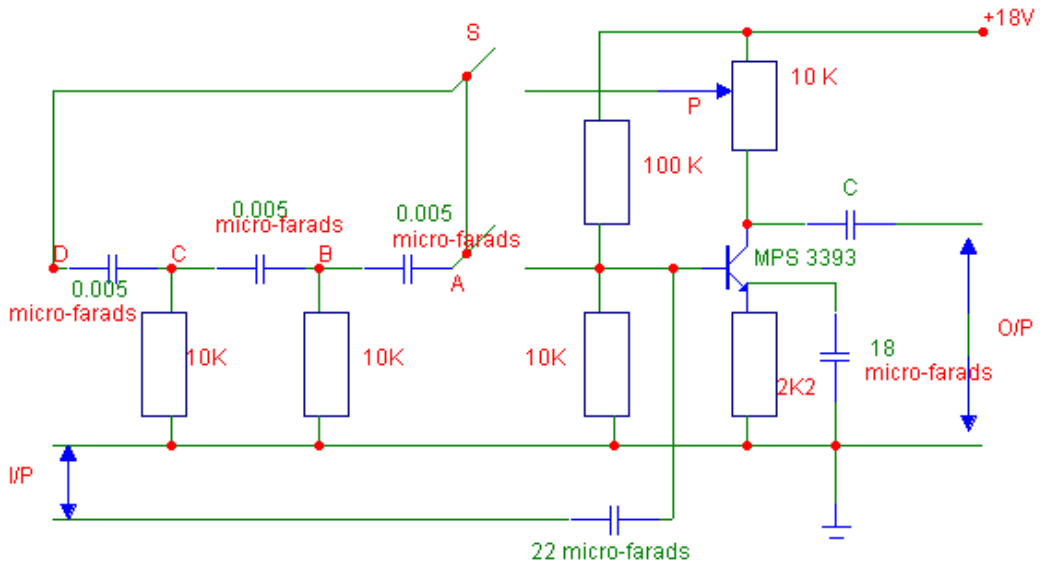


Fig. 1

2. With feedback in, adjust the gain until oscillation just occurs. Measure the frequency of oscillation on the CRO and compare with theory.

3. Measure the voltage amplitude at the output and at points D', C', B' & A'. Calculate the feedback ratio V_A/V_{out} and the attenuation factor V_A'/V_D' of the phase shift loop. Compare with theory.

4. Measure also the phase difference between the points D', C', B' & A'. This may be done by setting the channel 1 & 2 inputs to ground and adjusting the vertical shift until the two traces are coincident: then set the inputs both to a.c. and measure the period T and the interval ΔT between the two periodic traces. Then phase difference $\phi = (\Delta T/T) * 360°$.

5. Disconnect the feedback network and connect an a/c. input via the 22μF capacitor to the base of the transistor amplifier. Measure the amplifier gain and also the input and output impedances.

Part 2 – Colpitts Oscillator

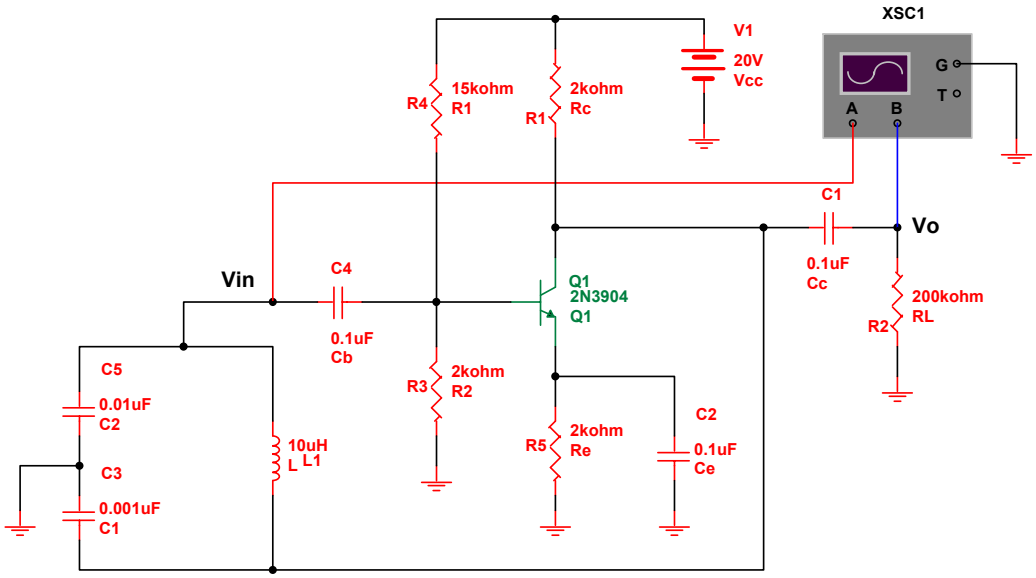
1. Set up the circuit of Colpitts oscillator as shown in Fig.2. Display the input (Vin) and output (Vo) waveforms on the scope and draw the resulting plots. Measure the time period (T) for one cycle of the output sine wave, peak-to-peak output voltage (Vo), peak-to-peak input voltage (Vin), and the phase difference between the input and output.

2. Based on the time period (T), determine the frequency of oscillation (f). Calculate the expected oscillator output frequency from the LC filter component values and compare. Also calculate the expected voltage gain based on the LC network capacitors C_1 and C_2, and compare it with the voltage gain from the input and output voltages.

3. Change the value of inductor L in the LC network to 5 μH. Record the time period (T), the output voltage (Vo) and the input voltage (Vin) and determine the frequency of oscillation. Calculate the expected oscillator output frequency based on the new value for L. Is the peak-to-peak output voltage affected? Explain.

4. Change the value of capacitor C_1 in the LC network to 500 pF. Again, record the period (T), the output voltage (Vo) and the input voltage (Vin) and determine the frequency of oscillation. Based on the new value of C_1, calculate the expected oscillator output frequency and explain any difference. Also calculate the voltage gain and explain any changes.

Figure 2: Colpitts Oscillator



C. Electrical Circuit Analysis – EE3304

ECA 1 – Circuit Analysis using PSpice

Purpose

1. To learn the basic features of PSpice
2. To use PSpice for the following
    - Basic DC Analysis
    - Thévenin/Norton equivalent of a circuit
    - Analysis of a circuit containing dependent sources

Introduction

The rapid advance in know-how, knowledge and expertise in the field of E&E engineering is paralleled by programs that use computers' increased capabilities in the solution of both traditional and novel problems. With the availability of tools for computer-aided circuit analysis, circuits of great complexity can be designed and analyzed within a shorter time and with less effort compared to the traditional methods. PSpice is a member of the SPICE (Simulation Program with Integrated Circuit Emphasis) family of circuit simulators. In the following exercises you will use PSpice to solve some circuits and to determine the quantities of interest. It is strongly suggested that you familiarize yourself with the basic PSpice tutorial provided on the our main server(einstein).

prelab

1. The circuit of Figure 1 contains a constant voltage source. Solve this circuit, using any method of your choice, and determine the current through all the elements and voltages of all nodes with respect to the ground.
2. Consider the circuit of Figure 2. Determine the Thévenin Equivalent of the circuit with respect to the nodes a,b. What value of resistance will you connect across this node so that it consumes maximum power? Determine the maximum power delivered to the load resistance R_L.
3. Figure 3 shows a circuit which contains two independent and two dependent sources. Use Mesh analysis and determine the **currents in all the branches**.

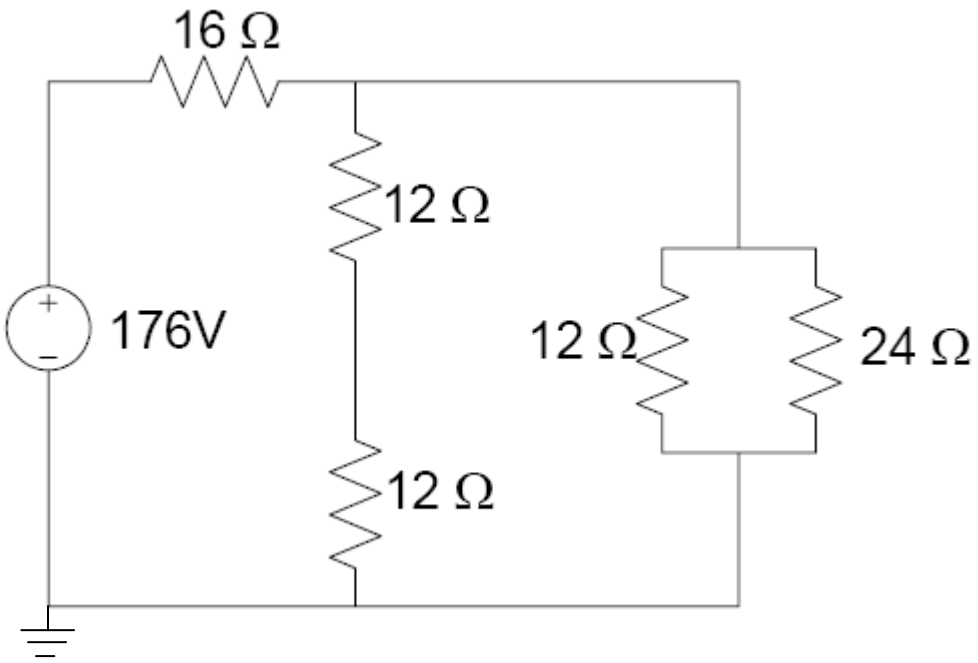


Figure 1 simple circuit

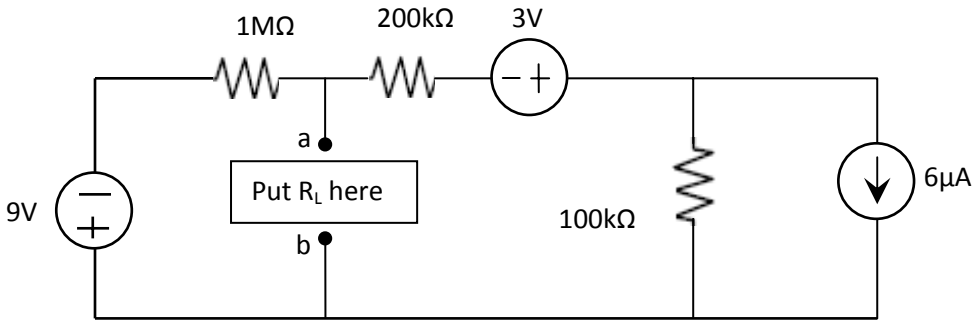


Figure 2 The original circuit for which the Thévenin equivalent is required in no. 2 of the Prelab.

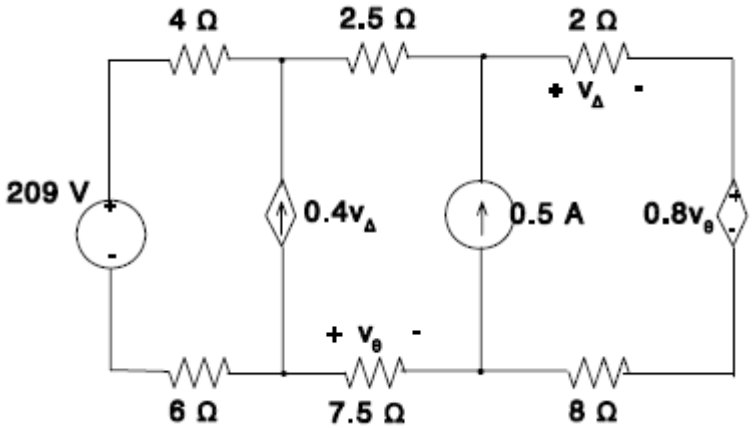


Figure 3 The circuit of no. 3

PSpice experiment

1. Perform DC analysis of the circuit of Figure 1 using PSpice. Display all node voltages and branch currents as explained in the PSpice Tutorial. Using the solution, verify that Kirchhoff's laws are satisfied in all the nodes and meshes.

2. Consider circuit of Figure 2

   2.1. Use PSpice to determine the Thévenin equivalent of the circuit with respect to nodes a,b. (Find the open circuit voltage and short circuit current).

   2.2. Consider that a resistance R_L (**i.e. create a global parameter for this resistor**) is connected across nodes a,b. Vary this resistance between 200k and 300k ohms in steps of 10k ohms as follows: use the DC sweep option of the "**PSpice>Edit Simulation Profile**" menu;. See the Fig. 4 for other settings in the menu. Run the simulation and use "**Trace>Add Trace**" to plot the variation of power absorbed by R_L. Use the "**Trace>Cursor>Display**" to determine the maximum power R_L consumes and the value of R_L which corresponds to this maximum power. Label this point on the plot.

3. Solve the circuit of Figure 3 using PSpice. Display all node voltages and branch currents. Verify that KVL is satisfied in all the meshes.

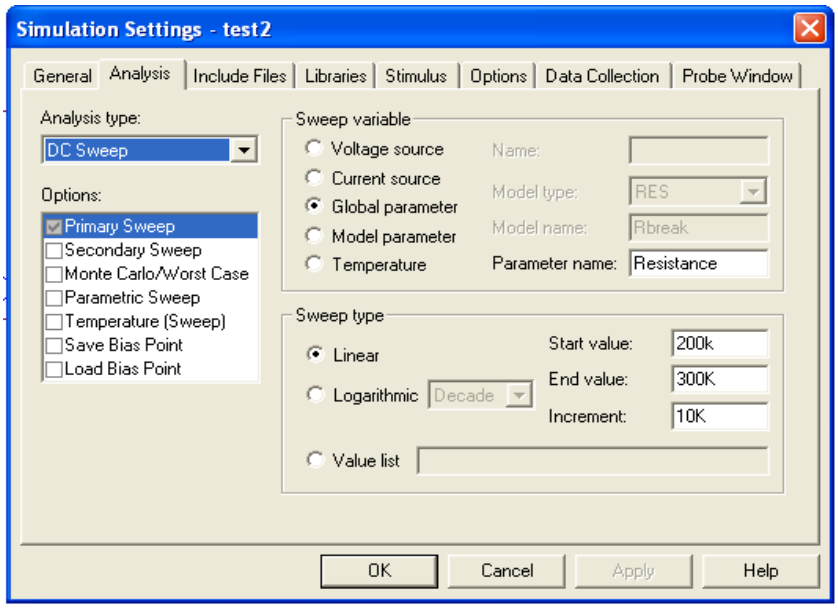


Figure 4: Settings for 2.

Report

1. Compare the results of the Prelab (Figure 1) with that given by PSpice.
2. Compare the results of the Prelab (Figure 2) with that given by PSpice. Based on your results (2.1 & 2.2) discuss how you can increase the amount of power delivered by a source. Is there any limit to the maximum power that can be delivered ?
3. Compare the results with that of Prelab (Figure 3).

For all the circuits, submit the Schematics, and any plots produced. In addition to the basic schematics, also provide the same diagrams with the node voltages and branch currents labelled.

ECA 2 – Step Responses of RL, RC & RLC circuits and Frequency Response & Resonance

Purpose

1. To study the response of RC, RL and RLC circuits when energized by an independent voltage source.
2. To determine the frequency response of RC and RLC circuits
3. To study resonance in a series RLC circuit excited by a sinusoidal source.

Introduction

Many phenomena that occur in electric circuits involve or produce time-dependent variables. When a *RC*, *RL*, or *RLC* circuit is suddenly energized or de-energized, a transient phenomenon, which dies out as the circuit approaches it steady-state operation, occurs. This is because of the way in which inductors and capacitors store energy and resistors dissipate it. The exact nature of the transients depends on the values of *R*, *L* and C as well as on how they are combined in a circuit. As you progress in your study of circuits you will meet instances where particular transient responses are beneficial and desired and, likewise, instances where such responses are detrimental to desired circuit behaviour.
The steady state response of the circuit, which is determined by the external source, is reached only after a transient time interval. The above describe the time domain solutions.

Likewise, the current and voltage responses in circuits containing inductors and capacitors (besides resistors) are frequency-dependent because the impedances of the former elements vary with frequency. Such circuits can be therefore used as "frequency-selective circuits" or filters - that is, they respond to signals of different frequencies differently and can therefore be designed to "pass" certain frequencies to a load or other parts of the circuit and to attenuate others. The study of the frequency response of a network involves the consideration of the magnitude and phase of the steady state voltages and currents as the frequency varies. An integral part of such a study involves the concept of resonance - a condition in which a circuit response can be maximized by adjusting either the circuit elements or the frequency of operation.

Prelab part I

1. Study tutorials on PSpice.
2. The circuit of Figure 1 is excited by a square wave voltage of 4V (peak or zero to peak) and which has a frequency of 100 Hz. In PSpice, use **VPULSE** to create this voltage waveform by setting V1 to 0 and V2 to 4. Use a pulse width of 5ms. (What is the pulse period here?) Use PSpice to determine the transient response (voltage across the capacitor) of the circuit. Limit your

transient response plot to one cycle of the source voltage. This may be done by setting the ”**Run to time**” parameter in the “**Edit Simulation**” window to be of the same period of the **VPULSE.** Obtain the response for the following two RC combinations:

$$R = 3.3\ \text{k}\Omega\ ;\ C = 0.1\ \mu\text{F};\ R = 3.3\ \text{k}\Omega\ ;\ C = 0.22\ \mu\text{F}$$

Calculate the time constant using the circuit parameters. Also, determine them from the plots. Remember the response changes by factor of 1/*e* during one time constant.

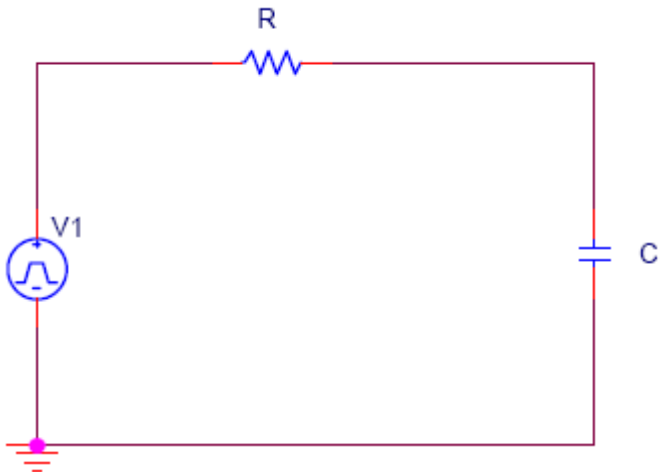


Figure 1 RC circuit with square wave

3. The circuit of Figure 2 is excited by a square wave voltage of 4V (peak to peak) having a frequency of 5 kHz. Again, create the waveform so that the pulse width is half the period. Use PSpice to determine the step response (voltage across the resistor) of the circuit. Limit your transient response plot to one cycle of the source voltage. Obtain the response for the following two *RL* combinations (simply change *R* and repeat):

   R=100 Ω ; *L*=1.0 mH; *R*=220 Ω ; *L*=1.0 mH

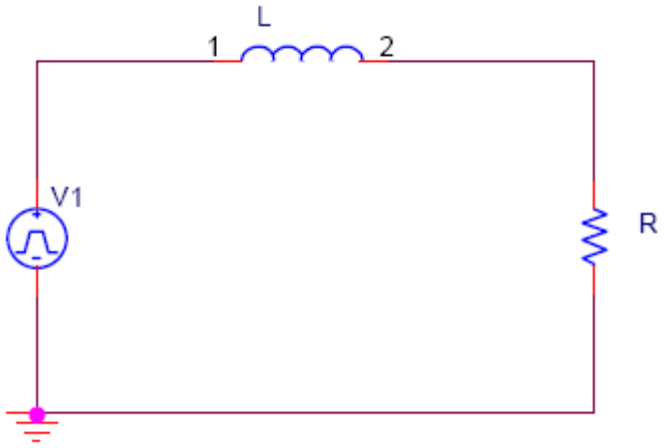


Figure 2 RL circuit with square wave.

Calculate the time constants using the circuit's parameters. Also, determine them from the plots. Remember the response changes by factor of 1/*e* during one time constant.

4. The circuit of Fig. 3 is excited by a symmetrical square wave voltage of 4V (peak) and frequency 500Hz. Using PSpice to obtain the plot of the variations of the voltage across the capacitor. Assume zero initial conditions. Limit the response to two cycles only. Do your simulations for the following two combinations.

R=10Ω, *L*=1mH, *C*=0.1μF; *R*=100Ω, *L*= 1mH, *C*=0.1μF;

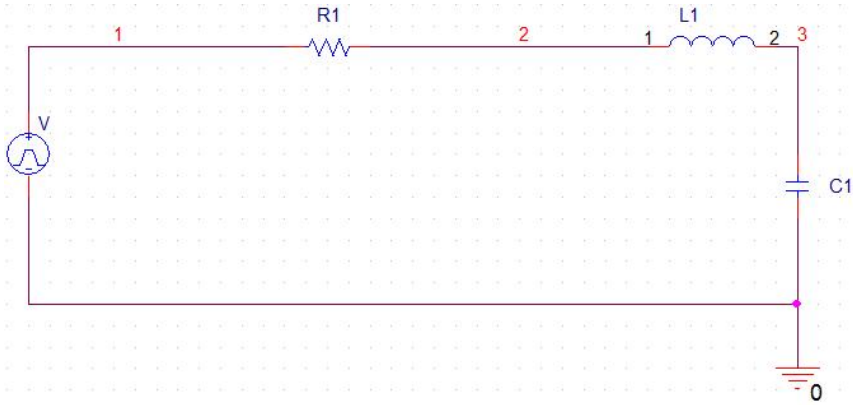


Figure 3 RLC circuit excited by a square wave

Find the frequency of the oscillations using the circuit parameters. Also, determine them from the plot. Note that, theoretically, this frequency is given by $\omega_0 = (1/LC)^{1/2}$.

Prelab Part II

Guidelines for Prelab: Draw schematics for the circuits indicated in Figures 1 and 2. Use source VSRC, AC=1. Since V_{in} =1, V_o itself will give the desired response.

Setup: AC Sweep, Linear and specify the frequency range of 10 Hz – 40kHz or as desired.

1. Let the circuit of Figure 1 be excited by a sinusoidal source whose frequency varies from 10Hz to 40 kHz. Use PSpice to obtain a plot of | vo/vin|.

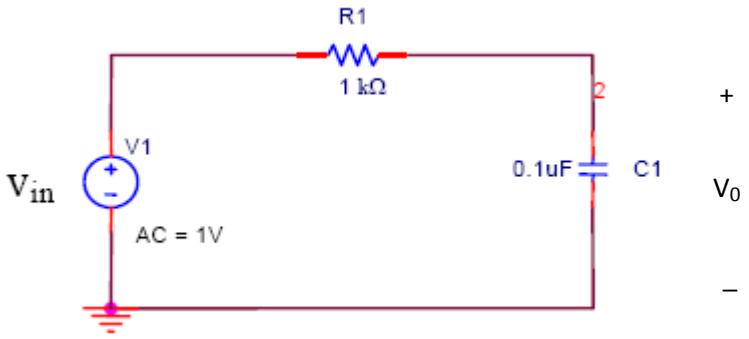


Figure 1 RC circuit

2. Consider the circuit of Figure 2. Let the circuit be excited by a sinusoidal source whose frequency can be varied form 100Hz to 100KHz. Use PSpice to obtain a plot of | v_0/v_{in} | for two different values of R1 (47 and 220 ohms). Obtain the two required plots on the same graph (by creating two circuits in the same Schematics file).

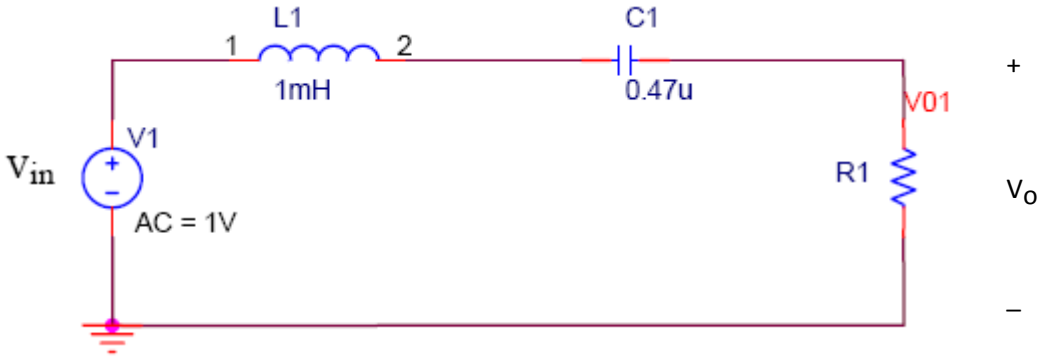


Figure 2 RLC series circuit

3. Calculate the resonant frequency and half-power frequencies of the circuit in Figure 3. Draw a phasor diagram (voltage across all elements at the resonant frequency).

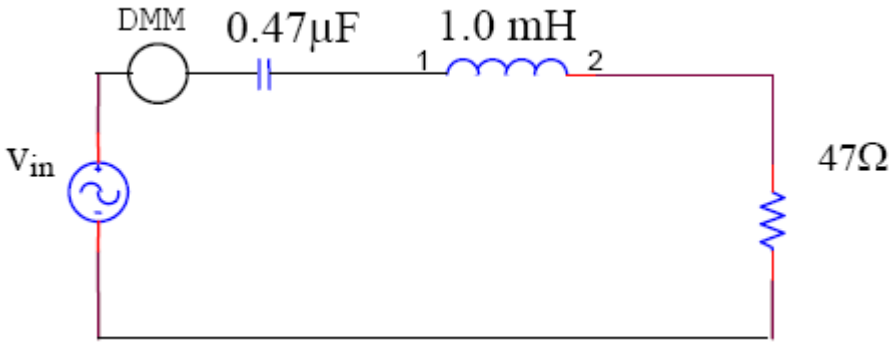


Figure 3 Circuit for part 3

Experiment

Part I

Set the controls on the function generator to output a square wave with an amplitude of 4V and a frequency of 100 Hz. Connect the output of the function generator to channel 1 of the scope. Adjust the scope to auto trigger and to use internal triggering off ch1. Why do we trigger the scope with the function generator output instead of triggering off the voltage across one of the circuit elements? There may be a trigger slope button on the scope; you can use this to view a different portion of the voltage trace.

Use the time base and amplitude controls to adjust the scope display. Likewise the vertical and horizontal positioning may be adjusted to move the curve. Don't forget to check the ground level of the scope to make sure amplitude measurements are accurate.

1. Construct the circuit of Figure 1. Adjust the function generator to provide 4 V (peak), 100Hz symmetrical square wave. Note that the use of this signal enables you to simulate the effect of repeatedly energizing the *RC* circuit and the waveform in the oscilloscope may be considered similar to the step response of this circuit. Sketch to scale the first half of the input square wave and voltage across the capacitor. Determine the time constant of the circuit by observing the waveform. Repeat the measurement for the second half of the input signal. Do your experiment for the following two combinations:

   R=3.3kΩ , C=0.1μF; R=3.3kΩ , C=0.22μF

2. Construct the circuit of Figure 2. Adjust the function generator to provide 4 V (peak), 5KHz symmetrical square wave. Sketch to scale the first half of the input square wave and the voltage across the resistor. Determine the time constant of the circuit. Do your experiment for the two combinations of *R* and *L* as in prelab.
3. Construct the circuit of Figure 3. Adjust the function generator to provide 4V (peak), 500Hz symmetrical square wave. Note that the use of this signal enables you to simulate the effect of repeatedly energizing the *RLC* circuit and the waveform in the oscilloscope may be considered similar to the step response of this circuit. Sketch to scale one half of the input square wave and the voltage across the capacitor. Do your experiment for the two combinations of the circuit elements as in prelab. For the response which is more oscillatory, try to estimate the frequency of oscillation.

Part II

1. Connect the circuit of Figure 1. Adjust the function generator to provide 3 V (6V peak-peak) supply. For the source frequency varying from 60Hz to 20kHz, measure v_0 and v_{in} for 12 different frequencies. Use the oscilloscope or counter to measure the frequency accurately.
2. Experimentally determine $| v_0/v_{in} |$ as a function of frequency for the circuit of Figure 2 for values of R1 equal to 47 and 220 ohms. Take more readings near the frequencies of interest. Use the DMM.
3. For the circuit of Figure 3 apply a 3 V (6V peak-peak) sinusoidal source. Vary the source frequency and experimentally determine the resonant frequency and half-power frequencies. Use the DMM to measure the current. Carefully note the point of resonance and measure the **magnitude and phase of all voltages (source & across all circuit elements)** at resonance using the oscilloscope. Make sure the source voltage is kept constant. Verify if the DMM can be used over the entire frequency range.

Report

General

Present all results in a clear and coherent manner. Use tables, graphs and diagrams as appropriate in addressing the particulars of the lab requirements listed below. Also, summarize your findings in a brief conclusion.

Part I

1. Based on your experiments with the two *RC* circuits, discuss the effect of change in *R*, *C* on the nature of the response and final value of the capacitor voltage. Compare the experimental plot with that obtained using PSpice (PreLab) and comment on the discrepancies.
2. Using your experiment with the *RL* circuits, discuss the effect of change in *R*, *L* on the nature of the response, final value of the resistor voltage and final value of the current in the circuit. Compare the experimentally determined time constant with that obtained using PSpice (Prelab) and comment on the discrepancies.
3. Compare the response of the *RLC* circuits obtained from the experiment with the PSpice simulation. Discuss the differences between the types of response for the two circuits. Compare the experimentally determined frequency of oscillation with the theoretical value.

Part II

1. Draw the frequency response of the *RC* circuit using your measurements. From your plot, determine the half-power frequency and compare it with the theoretical value. Compare the experimental plot with that obtained using PSpice. Suggest a possible application of this circuit.
2. Draw the frequency response for the series *RLC* circuit. Compare this plot with the plot obtained by the PSpice simulation. Discuss the influence of the value of the resistance in the frequency response of the circuit. Suggest an application for this circuit.
3. Compare the resonant frequency and bandwidth of the circuit based on your experiment with that of the Prelab. Compare the calculated and measured voltage phasors at resonance. Discuss the nature of the circuit at resonance using a suitable phasor diagram based on your experiment.

ECA 3 – Design of a simple filter

Purpose

1. To design a filter to pass a signal at a desired frequency and reject the signal at another frequency.
2. To simulate this circuit in PSpice and study its frequency response.
3. To experimentally verify the performance of the filter.

Introduction

A filter is a frequency-selective device which allows certain frequencies to pass through it while blocking or severely limiting others. It is possible to design simple circuits using resistance, inductance and capacitance which act as filters. Here a study the frequency response of a particular *RLC* circuit configuration is examined.

Prelab

1. Consider the circuit of Figure 1. The circuit capacitors C1 and C2 are to be chosen so that the circuit effectively transmits a 25-kHz signal and effectively blocks a 50-kHz signal.
    - Calculate C2: The L1 and C2 parallel combination should be an open circuit at 50 kHz (i.e. choose C2 so that the impedance of the parallel combination is infinite).
    - Calculate C1: Use the value of C2 determined above and minimize the overall impedance of the circuit at 25 kHz.
2. In PSpice, simulate the circuit of Figure 1 using a sinusoidal source (VSRC with AC=1) whose frequency varies from 10kHz to 60 kHz. The simulation should be set up to do an AC sweep. Obtain a decibel plot of |V0/Vin| (see Figure 2). Use the 'cursor' features of PSpice to show the response of this circuit at the frequencies of interest.

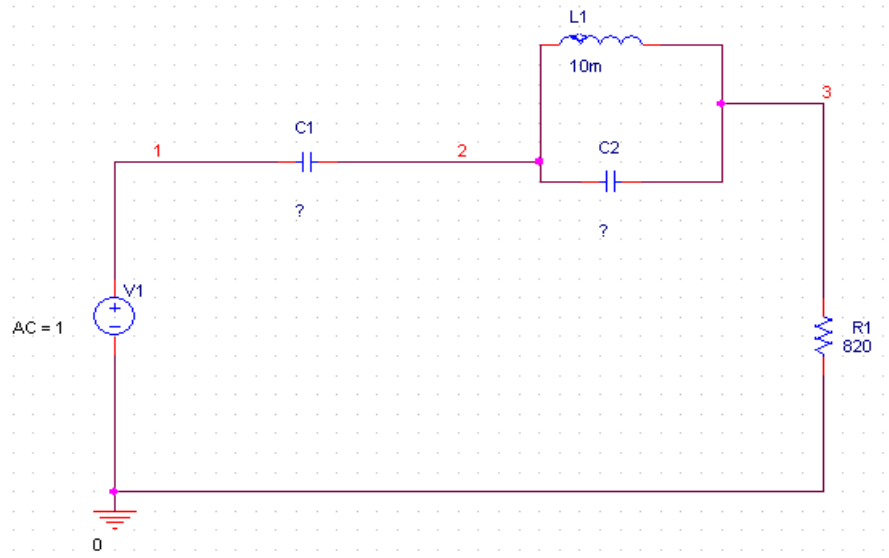


Figure 1 Frequency-Selective Circuit

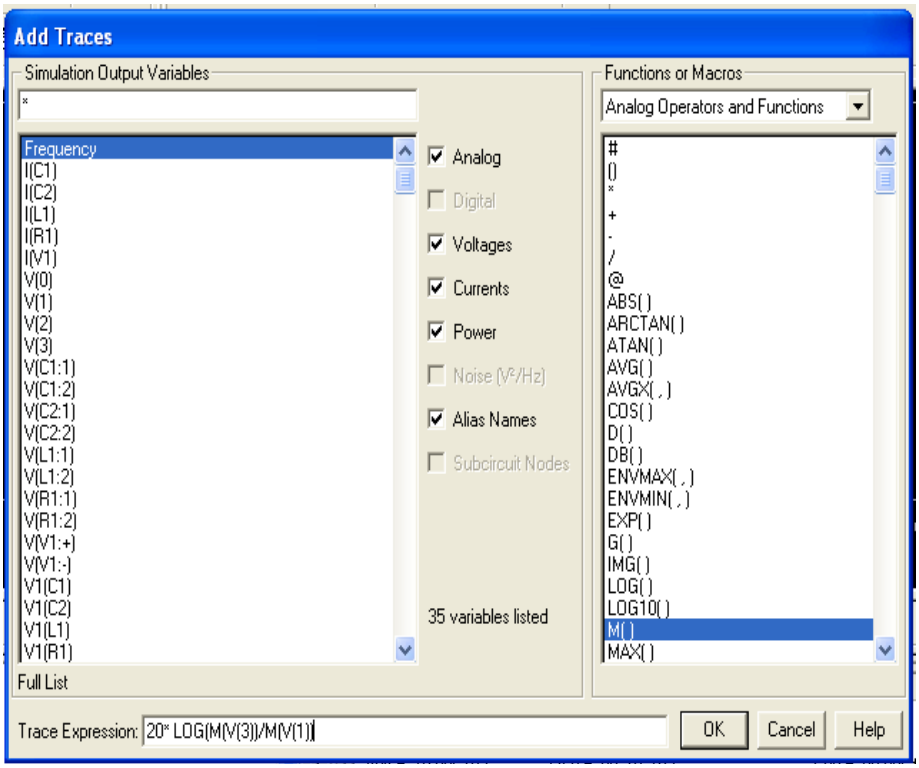


Figure 2 Obtaining a decibel plot of the output/input magnitude ratio

Experiment

Construct the circuit of Figure 1. Use the components based on the Prelab. Adjust the input for 5 V (peak; or 10 V peak to peak) sinusoidal. For the source frequency varying from 10kHz to 60 kHz, measure V0 and Vin for 20 different frequencies using Oscilloscope. In particular, carefully record the output voltages near the frequencies of interest (20 kHz and 50 kHz).

Report

Plot the frequency response of the circuit you have designed based on your measurements. Compare the experimental plot with that obtained using PSpice. Discuss your results.

D. Analogue Communications – EE3308

ACM 1 – Fourier Theory

OBJECTIVES

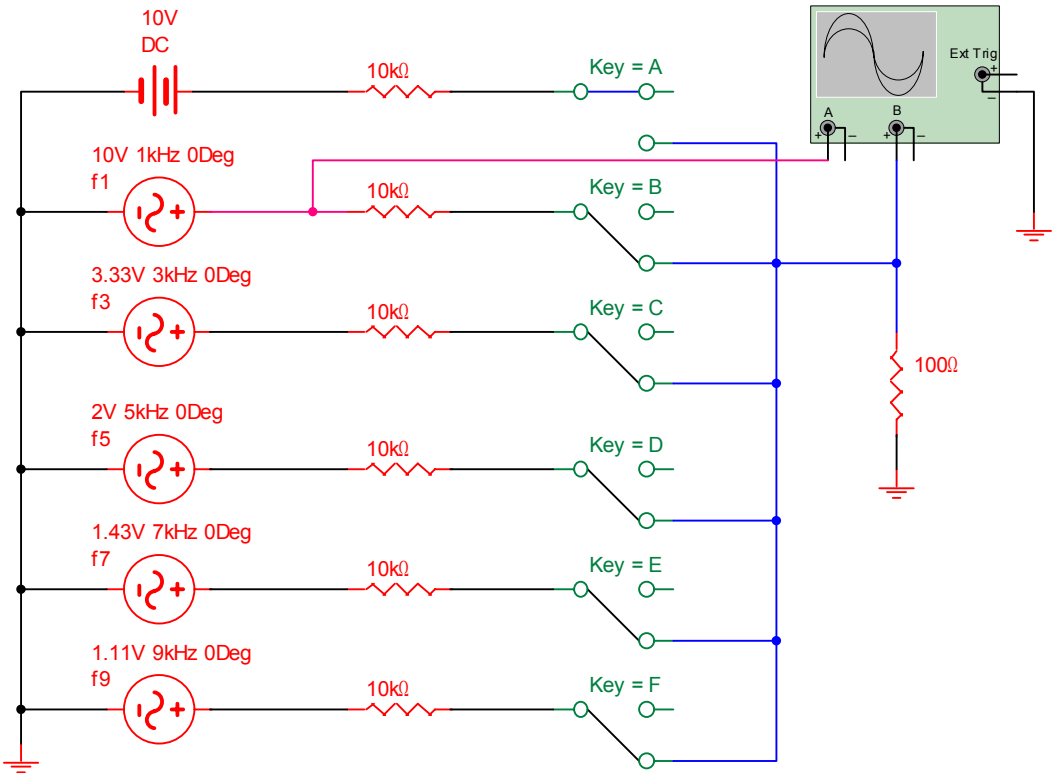
1. Learn how a square wave and a triangular wave can be produced from a series of sine waves and cosine waves, respectively, at different frequencies and amplitudes.
2. Learn about the difference between curve plots in time and frequency domains and examine periodic pulses with different duty cycles in both domains.

EQUIPMENT

This experiment will use the Electronics Workbench (Multisim 9) simulation.

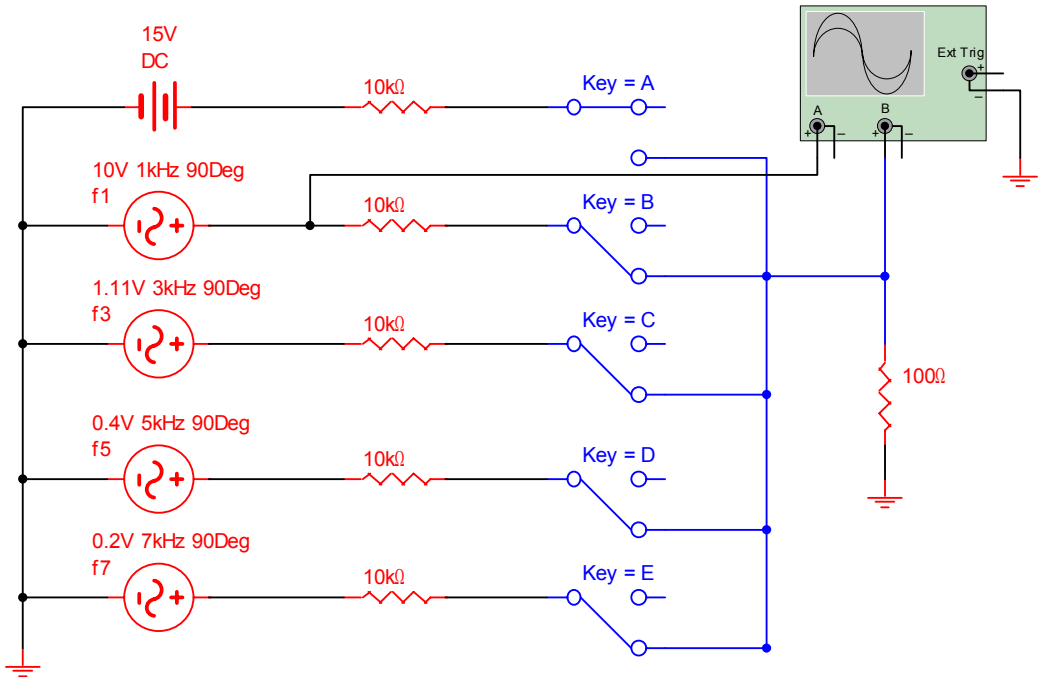
PROCEDURE

1. Draw the circuit shown below. Make sure that the following oscilloscope settings are selected: Time base (Scale = 200 us/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = 0, DC), Ch B (Scale = 50 mV/Div, Ypos = 0, DC), Trigger (Pos edge, Level = 0, Auto). You will generate a square wave curve plot on the oscilloscope screen from a series of sine waves called a Fourier series. Run the simulation and draw the square wave curve and the fundamental sine wave curve on the same plot. Measure the period (T) for the square wave and the sine wave and calculate their frequencies. What is the relationship between the fundamental sine wave frequency and the square wave frequency, the frequencies of sine wave generators (f1, f3, f5, f7 and f9)? What is the relationship between the amplitude of each of the harmonic sine waves and the amplitude of the fundamental sine wave?



2. Press the A key to close switch A to add a dc voltage level to the square wave curve plot and run the simulation again. Change the scope settings as needed. Draw the new square wave curve plot and explain what happened to the plot. Press the F and E keys to open switches F and E to eliminate the ninth and seventh harmonic sine waves. Run the simulation again and draw the new curve plot. Note any change on the graph. Press D key to eliminate the fifth harmonic sine wave and run the simulation again. Draw the new curve plot and note any change on the graph. Eliminate the third harmonic (switch C) and run the simulation again. Explain what happened to the square wave curve plot.

3. Draw the circuit shown below. Make sure that the following oscilloscope settings are selected: Time base (Scale = 200 us/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = 0, DC), Ch B (Scale = 100 mV/Div, Ypos = 0, DC), Trigger (Pos edge, Level = 0, Auto). You will generate a triangular wave curve plot on the oscilloscope screen from a series of cosine waves. Run the simulation and draw the triangular wave curve and the fundamental cosine wave curve on the same plot. Calculate the frequency (f) of the triangular wave from the period (T). What is the relationship between the fundamental frequency and the triangular wave frequency, the frequencies of generators (f1, f3, f5, and f7)? What is the relationship between the amplitude of each of the harmonic generators and the amplitude of the fundamental generator?



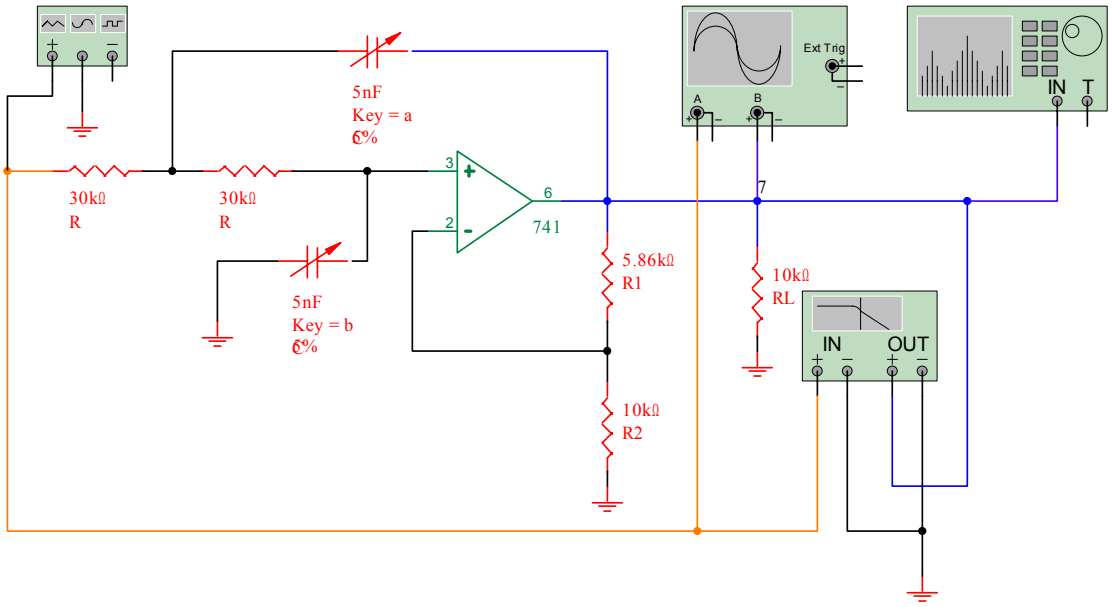
4. Press the A key to close switch A to add a dc voltage level to the triangular wave curve plot and run the simulation again. Draw the new triangular wave curve plot and explain what happened to the plot. Eliminate the seventh and fifth harmonic waves. Run the simulation again and draw the new curve plot. Note any change on the graph. Eliminate the third harmonic wave and run the simulation again. Draw the new curve plot and note any change on the graph. Explain what happened to the triangular wave curve plot.

5. Draw the circuit shown below. Make sure that the following function generator settings are selected: *Square wave*, Freq = 1 kHz, Duty cycle = 50%, Ampl = 2.5 V, Offset = 2.5 V. Make sure that the following oscilloscope settings are selected: Time base (Scale = 500 us/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = 0, DC), Ch B (Scale = 5 V/Div, Ypos = 0, DC), Trigger (Pos edge, Level = 0, Auto). You will plot a square wave in time domain at the input and output of a two-pole low-pass Butterworth filter.

6. Bring down the oscilloscope enlargement and run the simulation to one full screen display, then pause the simulation. Draw the displayed curve plots. The channel A curve is the filter input (5) and channel B curve is the filter output (7). Are the curve plots the same shape, disregarding the amplitude differences? Use the cursors to measure the period (T) and the up time (t_o) of the input curve and calculate the pulse duty cycle.

7. Bring down the Bode plotter enlargement to display the Bode plot of the filter. Make sure that the following settings are selected: *Magnitude*, Vertical *(Log, F = 10*

dB, I = -40 dB), Horizontal (*Log, F = 200 kHz, I = 100 Hz*). Run the simulation to completion and use the cursor to measure the cut-off frequency (fc) of the low-pass filter.



8. Bring down the spectrum analyser enlargement and make sure that the following settings are selected: Freq (Start = 0 kHz, Center = 5 kHz, End = 10 kHz), Ampl (Lin, Range = 1 V/Div), Res = 50 Hz. Run the simulation until the resolution frequencies match, then pause the simulation. Draw the displayed filter output square wave frequency spectrum and record the amplitude of each spectral line in a table shown below.

	Frequency (kHz)	Amplitude
f1	1	
f2	2	
f3	3	
f4	4	
f5	5	
f6	6	
f7	7	
f8	8	
f9	9	

What conclusions can you draw about the difference between the even and odd harmonics for a square wave with the duty cycle calculated in step 6?

What conclusion can you draw about the amplitude of each odd harmonic compared to the fundamental for a square wave with the duty cycle calculated in step 6?

Was this frequency spectrum what you expected?

Based on the filter cut-off frequency measured in step 7, how many of the square wave harmonics would you expect to be passed by this filter? Based on this answer, would you expect much distortion of the input square wave at the filter output? Explain.

ACM 2 – Amplitude Modulation

OBJECTIVES

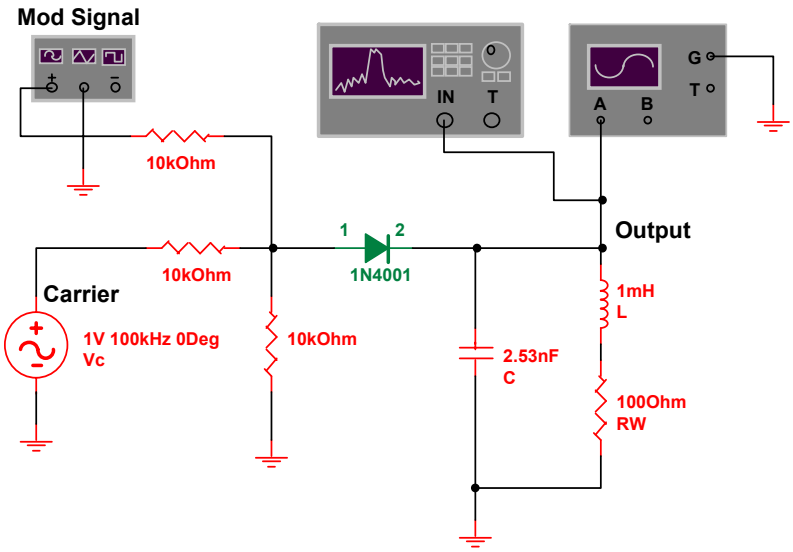
1. Demonstrate an amplitude-modulated carrier in the time and frequency domains for different modulation indexes and modulating frequencies.
2. Determine the modulation index, percent modulation, signal bandwidth and side frequencies of amplitude-modulated carrier.

EQUIPMENT

One DC Voltage supply
Two Function Generators
One Spectrum analyser
One dual-trace oscilloscope
One 1N4001 Diode
Capacitor: 2.53 nF
Inductor: 1 mH
Resistors: 100, 2K, 10K, 20K

PROCEDURE

AM DIODE MODULATOR



1. Set up the circuit of figure above. The modulating signal (sine wave) frequency should be 5 kHz, with amplitude 1 V. Set the carrier frequency to 100 kHz and amplitude to 1 V. Display an amplitude-modulated carrier on the scope and draw the

plot. Also calculate the modulation index and percent modulation based on baseband (V_m) and carrier (V_c) amplitudes. Compare them with the ones determined from the plot.

2. Display the frequency spectrum of the modulated carrier on the spectrum analyser. Draw the spectral plot. Measure the carrier frequency (f_c), the upper side frequency (f_{usf}), the lower side frequency (f_{lsf}), and the voltage amplitude of each spectral line and show them on the spectral plot. Determine the bandwidth from the plot.

3. Change the modulating signal amplitude to 0.5 V and repeat step 1 above. How did the values of the modulation index and percent modulation compare with the initial ones?

4. Display the frequency spectrum of the new modulated carrier on the spectrum analyser. How do the side frequencies' amplitudes and the bandwidth from this frequency spectrum compare with the ones in step 2 above?

5. Change the modulating signal amplitude to 0 V and repeat step 1 above. How did the wave shape compare with the previous amplitude-modulated waveforms? Explain any difference.

6. Draw the new spectral plot and compare it with the previous ones.

7. Change the modulating frequency to 10 kHz and the amplitude back to 1 V. Again display the amplitude-modulated carrier on the scope and draw the plot. How do the waveform differ from the waveform for a 5 kHz modulating frequency in step 1?

8. Connect the spectrum analyser and repeat step 2 for the 10 kHz modulating signal. How do the bandwidth of the modulated carrier for a 10 kHz modulating frequency compare with the bandwidth for a 5 kHz modulating frequency? Was the any difference between the amplitude of the side frequencies for the 10 kHz plot and the 5 kHz plot? Explain.

9. Change the modulating signal frequency back to 5 kHz and select square wave on the function generator. Display the carrier modulated by a square wave on the oscilloscope screen. Draw the plot. How do the wave shape differ from the wave shape in step 1? Determine the modulation index and percent modulation from the plot.

10. Use the spectrum analyser to display the spectrum for the square wave modulated carrier. Draw the spectral plot and ignore any side frequencies with amplitudes less than 10% of the carrier amplitude. Measure the frequency of the spectral lines and record answers on the plot. How do the frequency spectrum for the square-wave modulated carrier differ from the spectrum for the sine wave

modulated carrier? Explain why they are different. Determine the bandwidth of the modulated from the spectral plot and compare it with the bandwidth for the 5-kHz sine wave modulated carrier?

ACM 3 – Frequency Modulation

OBJECTIVES

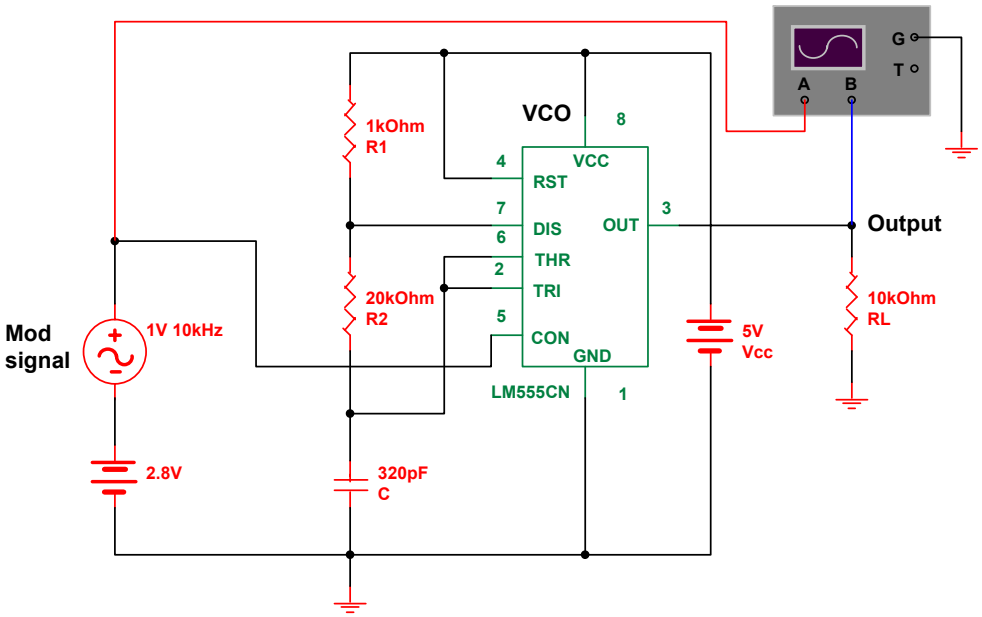
1. Demonstrate a frequency-modulated carrier in the time domain and frequency domain for different modulation indexes and modulating frequencies.
2. Learn how to determine the modulation index of a frequency-modulated carrier using the time and frequency domains displays.
3. Learn how to use the Bessel table and determine the signal bandwidth

EQUIPMENT

One Function generator
One Oscilloscope
One Spectrum analyser
Two dc voltage supplies
One LM555 timer
Capacitor: 320 pF
Resistors: 1K, 10K, 20K

PROCEDURE

The following steps should be performed using the modulator circuit shown below.



1. Display the frequency-modulated carrier on the oscilloscope screen. Draw the curve plot. Measure the time period for one cycle (T) of the lowest frequency and the highest frequency and record your answers on the plot.

2. Based on the time periods (T_1 and T_2), calculate the lowest and highest frequencies (f_1 and f_2) and then determine the frequency deviation (δ). Based on δ and the modulating frequency (f_m), calculate the modulation index (m_f) for the frequency-modulated carrier.

3. Connect the spectrum analyser and display the frequency spectrum for the frequency-modulated carrier. Draw the spectral plot. Measure the amplitude of each spectral line and compare these amplitudes with the expected values from the Bessel table for the modulation index determined in step 2. [N.B. The spectrum analyser will display both negative and positive spectral lines as positive.]

4. Measure the frequency difference between the spectral lines and compare it with the modulating signal frequency. Determine the bandwidth of the modulated carrier from the spectral plot, ignoring the spectral lines with amplitudes less than one percent of the unmodulated carrier amplitude of 1 V. Compare this value with the bandwidth calculated using Carson's rule. Also compare this value with the bandwidth calculated using the modulation index and Bessel table. Which method is more accurate? Explain why.

5. Change the modulating frequency to 20 kHz and repeat steps 1 to 4. How did the bandwidth of this modulated carrier compare with the bandwidth measured for a modulating frequency of 10 kHz? Explain any difference.

E. Digital Communications – EE3309

DCM 1 – FDM and TDM

OBJECTIVES

1. Demonstrate how frequency-division multiplexing makes it possible to simultaneously transmit many analog signals on one communications channel.
2. Demonstrate how time-division multiplexing makes it possible to simultaneously transmit more than one digital signal on one communications channel.

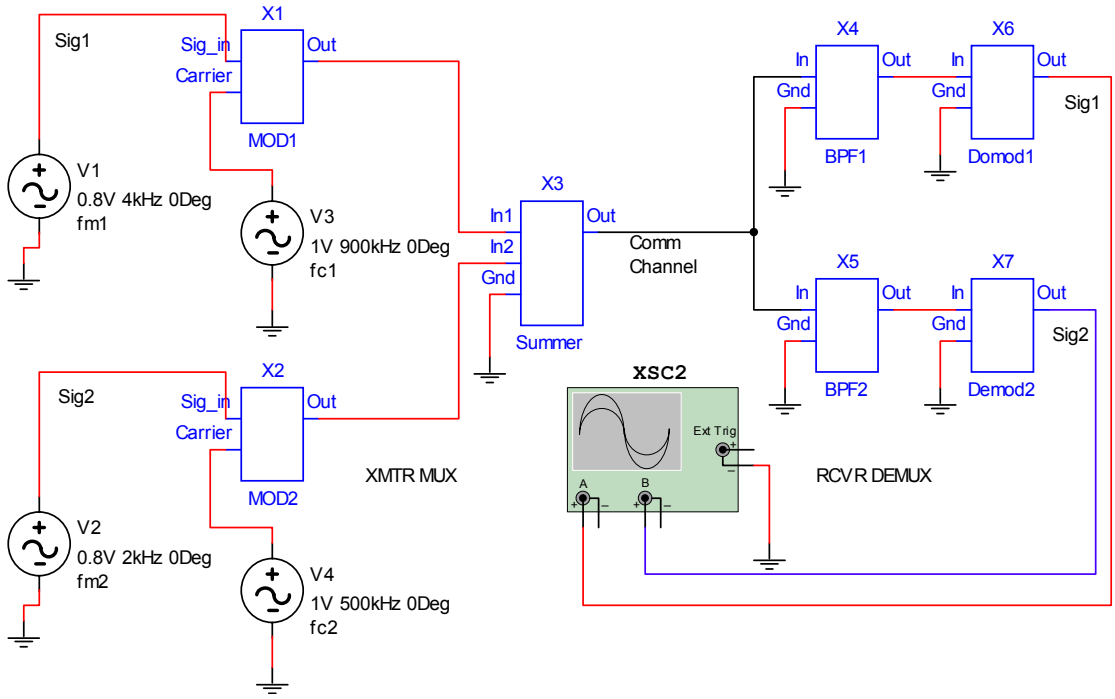
EQUIPMENT

This experiment will use the Electronics Workbench (Multisim 9) simulation.

PROCEDURE

Part A – FDM

1. Draw the circuit shown below. The sub-circuits for the demodulators, filters, modulators and the summer are provided within the files named DCM1-Demod, DMC1-BPF, DCM1-MOD, and DCM1-SUMMER respectively. Bring down the oscilloscope enlargement. Make sure that the following settings are selected: Time base (Scale = 100 us/Div, Xpos = 0, Y/T), Ch A (Scale = 2 V/Div, Ypos = 1.0, DC), Ch B (Scale = 2 V/Div, Ypos = -2.0, DC), Trigger (Pos edge, Level = 0, Auto). Run the simulation to completion. Draw the curve plots for the FDM receiver Channel 1 output and receiver Channel 2 output as displayed on the scope. Measure the period (T) of the receiver Channel 1 output, calculate the output frequency (f1) and compare it with transmitter Channel 1 input frequency (fm1). Also measure the period of the receiver Channel 2 output, calculate its frequency (f2) and compare it with the transmitter Channel 2 input frequency (fm2). What can you conclude about the tuned frequencies of band-pass filters BPF1 and BPF2? Why are the output sine waves slightly distorted?



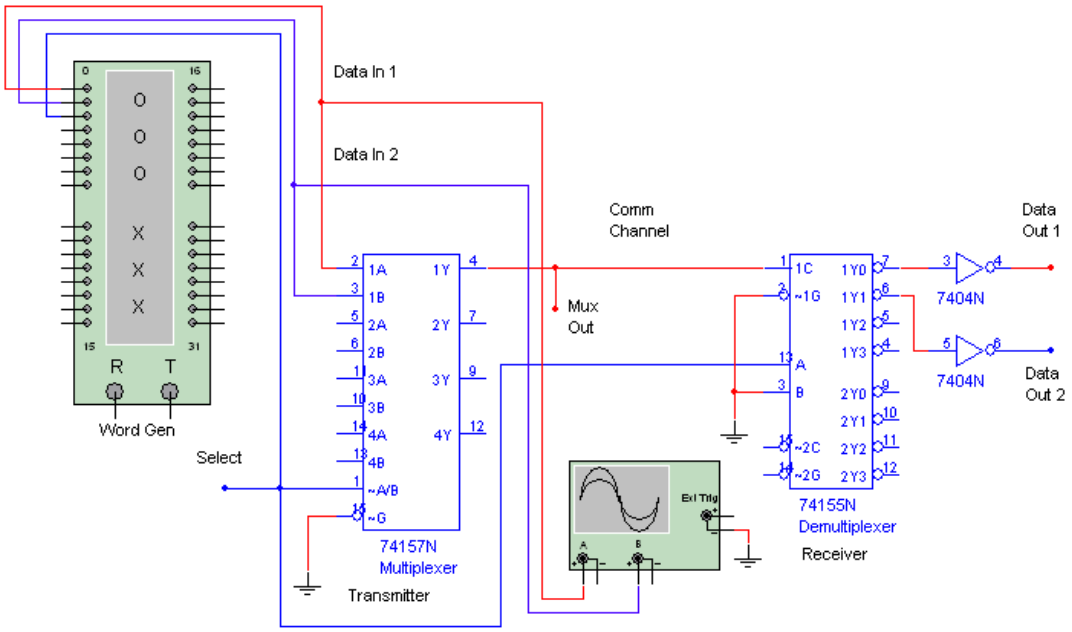
2. Reduce the Channel 2 input signal source (V2) to 0 V. Run the simulation to completion, draw the scope display, record and explain what happened to the receiver Channel 2 output. Is the receiver Channel 1 output affected? Explain why. Restore the Channel 2 input and reduce the Channel 1 input source (V1) to 0 V. Run the simulation to completion, draw the scope display, record and explain what happened to the receiver Channel 1 output.

3. Connect the oscilloscope Input A to the output of BPF1 and Input B to the output of BPF2. Bring down the oscilloscope enlargement and change the Time base scale to 1 us/Div. Run the simulation to ten full screen displays, then pause the simulation. Draw the curves displayed on the scope and measure the period of the receiver BPF1 output (top curve plot) and calculate its frequency (fc1). How does the BPF1 output frequency compare with the carrier frequency feeding modulator 1 on the transmitter? Measure the period of the receiver BPF2 output (bottom curve plot) and calculate its frequency (fc2). How does the BPF2 output frequency compare with the carrier frequency feeding modulator 2 on the transmitter? Were these expected? Explain why?

4. Connect the oscilloscope Input A to the communications channel. Bring down the oscilloscope enlargement and change Channel A scale to 5 V/Div. Run the simulation to ten full screen displays, then pause the simulation. Draw the composite waveshape transmitted by the transmitter multiplexer. Was the composite waveshape a sine wave? If not, explain why.

Part B – TDM

5. Draw the circuit shown below and double click the word generator and make sure the following settings are selected: Controls (Cycle), Display (Hex), Trigger (Internal, Pos edge) and Frequency (1 kHz). Make sure the right hand columns of the first 16 rows of words are respectively entered as 0, 0, 1, 1, 0, 0, 1, 1, 6, 4, 6, 4, 6, 4, 6, 4. Bring down the oscilloscope enlargement and make sure that the following settings are selected: Time base (Scale = 2 ms/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = 1.0, DC), Ch B (Scale = 5 V/Div, Ypos = -2.0, DC), Trigger (Pos edge, Level = 0, Auto). Run the simulation and pause it after one full oscilloscope screen display. Draw the curve plots of the multiplexer Data Input 1 and Data Input 2 as shown on the screen. If the bit time is 1 ms, what is the serial 8-bit binary code at the multiplexer Data Input 1 and Data Input 2? Are both 8-bit binary codes being input to the multiplexer at the same time? Explain why?



6. Connect the oscilloscope Channel A input to the multiplexer output (Mux Out) and Channel B input to the multiplexer select input. Run the simulation and pause it after one full scope screen display. Draw the plot for the composite signal transmitted by the multiplexer (top curve) and the multiplexer select input (bottom curve). Does the composite signal consist of both 8-bit binary input codes? Was this expected? From the curve plots, determine the period for one TDM frame. What is the relationship between the period for one TDM frame and the total time required to transmit both multiplexer data inputs one time? What is the relationship between the multiplexer select input and the composite signal transmitted by the multiplexer?

7. Connect the oscilloscope Channel A input to the multiplexer Data Input 1 and Channel B input to the demultiplexer Data Output 1. Run the simulation and pause it after one full scope screen display. Draw the plot for the multiplexer Data Input 1 (top curve) and the demultiplexer Data Output 1 (bottom curve). How does the demultiplexer Data Output 1 signal compare with the multiplexer Data Input 1? Repeat step 7 for the multiplexer Data Input 2 and the demultiplexer Data Output 2.

8. Connect the oscilloscope Channel A input to the demultiplexer Data Output 1 and Channel B input to the demultiplexer Data Output 2. Run the simulation and pause it after one full scope screen display. Draw the plot for the demultiplexer Data Output 1 (top curve) and the demultiplexer Data Output 2 (bottom curve). How does the demultiplexer Data Output 1 signal compare with the demultiplexer Data Output 2 signal? What conclusion can you draw about what the demultiplexer is doing?

DCM 2 – Digital Encoding and PCM

OBJECTIVES

1. Learn about serial data transmission, binary encoding, channel capacity and the effect of multilevel encoding on bit rate.
2. Demonstrate PCM encoding and decoding using ADC and DAC.

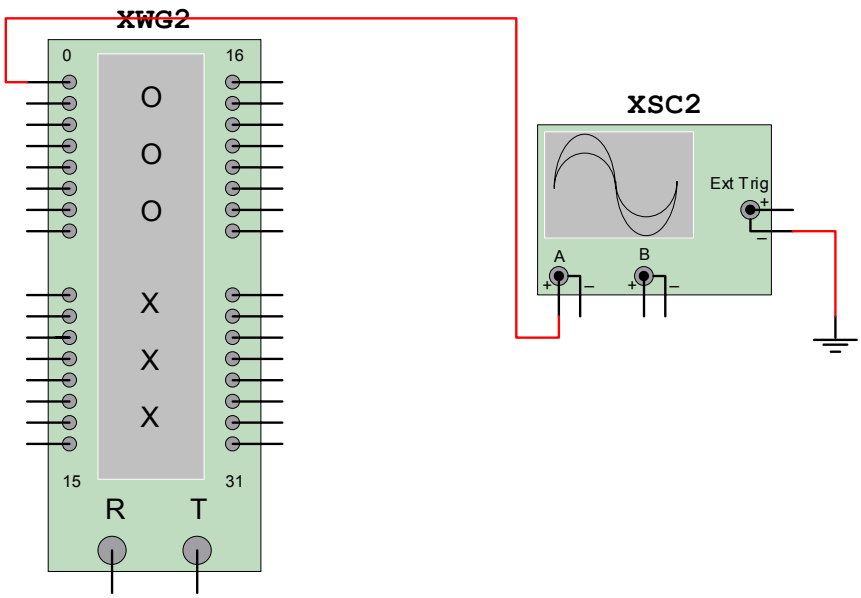
EQUIPMENT

This experiment will use the Electronics Workbench (Multisim 9) simulation.

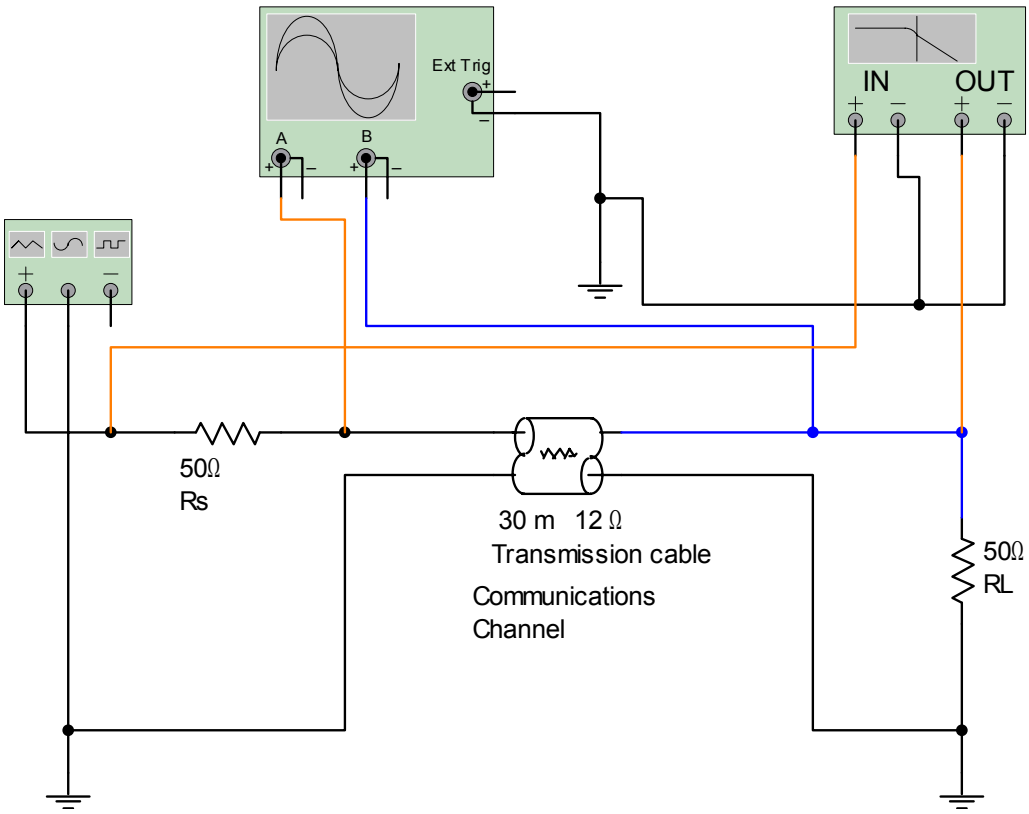
PROCEDURE

Part A – Digital Encoding

1. Draw the circuit shown below and double click the word generator and make sure the following settings are selected: Controls (Burst), Display (Hex), Trigger (Internal, Pos edge) and Frequency (1 MHz). Make sure the right hand columns of the first 10 rows of words are respectively entered as 1, 0, 1, 0, 0, 0, 1, 1, 0, 1.Remember to set the word as the final position. Make sure that the following oscilloscope settings are selected: Time base (Scale = 1 us/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = 0, DC), Trigger (Pos edge, Level = 0, Auto). Double click the word generator to bring down the enlargement and click burst to run the simulation. On the scope screen you are looking at a serial asynchronous transmission of a 7-bit ASCII code with a start bit before the beginning of the ASCII code, a stop bit after the end of ASCII code, and a bit time (t_b) of 1 µs. A mark is being transmitted during the idle time at the beginning of the transmission. Draw the plot as displayed on the scope. What character (number or letter) does the 7-bit ASCII code represent? [*Remember that the least significant bit is transmitted first, after the start bit*.] Based on the bit time (t_b), determine the bit rate in bits per second (bps) and the baud rate for the displayed signal. Compare the bit rate and the baud rate for this digital signal.

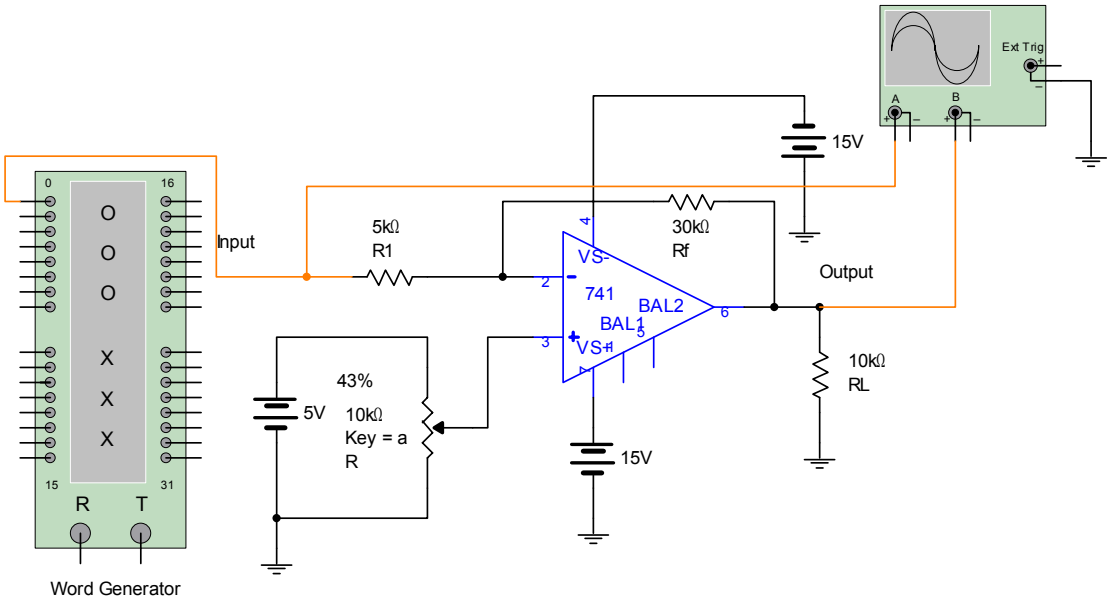


2. Wire the circuit below and bring down the oscilloscope enlargement. Make sure that the following settings are selected: Time base (Scale = 5 us/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = 0, DC), Ch B (Scale = 1 V/Div, Ypos = 0, DC), Trigger (Pos edge, Level = 0, Auto). Double click the function generator to bring down the enlargement. Make sure the following settings are selected: Waveform = Square Wave, Freq = 100 kHz, Duty Cycle = 50%, Ampl = 2.5 V, Offset = 2.5 V. Also bring down the Bode plotter enlargement and check the following settings: Magnitude, Vertical (Log, F = -15 dB, I = -35 dB), Horizontal (Log, F = 10 MHz, I = 10 kHz). Run the simulation and pause it after two full screen displays on the scope. Draw the cable input and output curve plots from the scope. Measure and record the cable bandwidth (BW) on the Bode plotter. Assuming that alternate ones and zeros are being transmitted (10101010----), determine the cable input bit time (t_b) and calculate the bit rate.

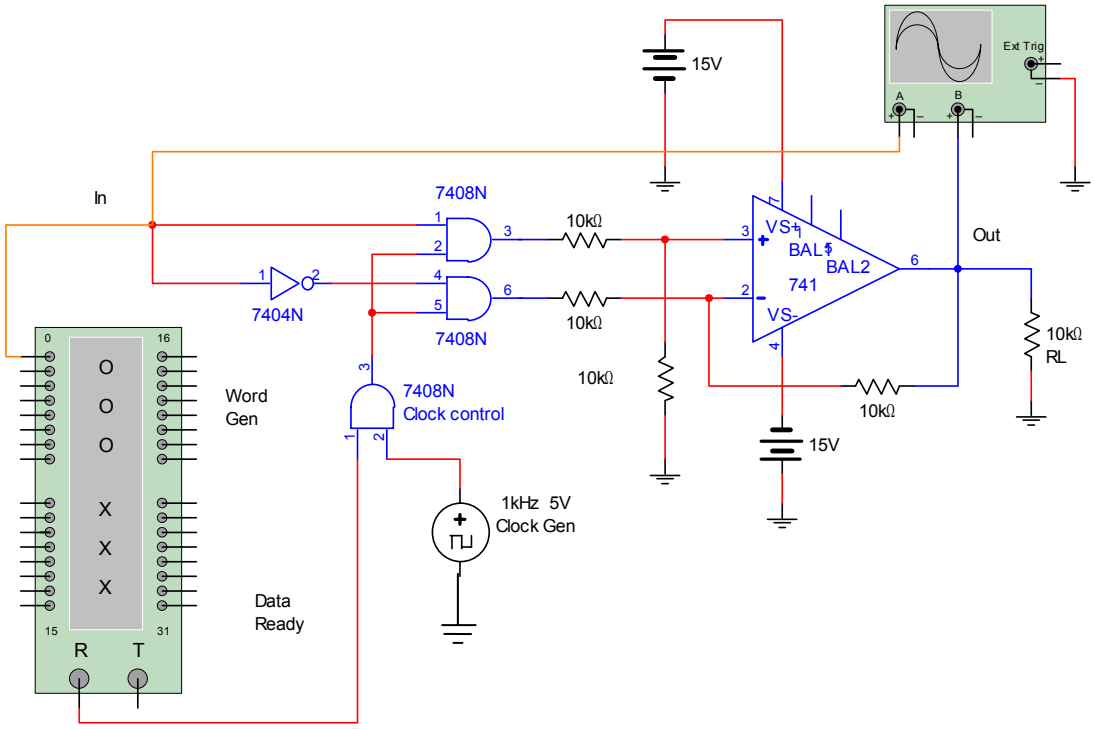


3. Change the square wave frequency on the function generator to 1 MHz and the scope time base scale to 500 ns/Div and run the simulation as above. Determine the new bit time and bit rate. What is the relationship between the bit rate and the cable bandwidth? How does this bit rate compare with the theoretical channel capacity (C)? Is the cable output square wave the same shape as the cable input square wave?

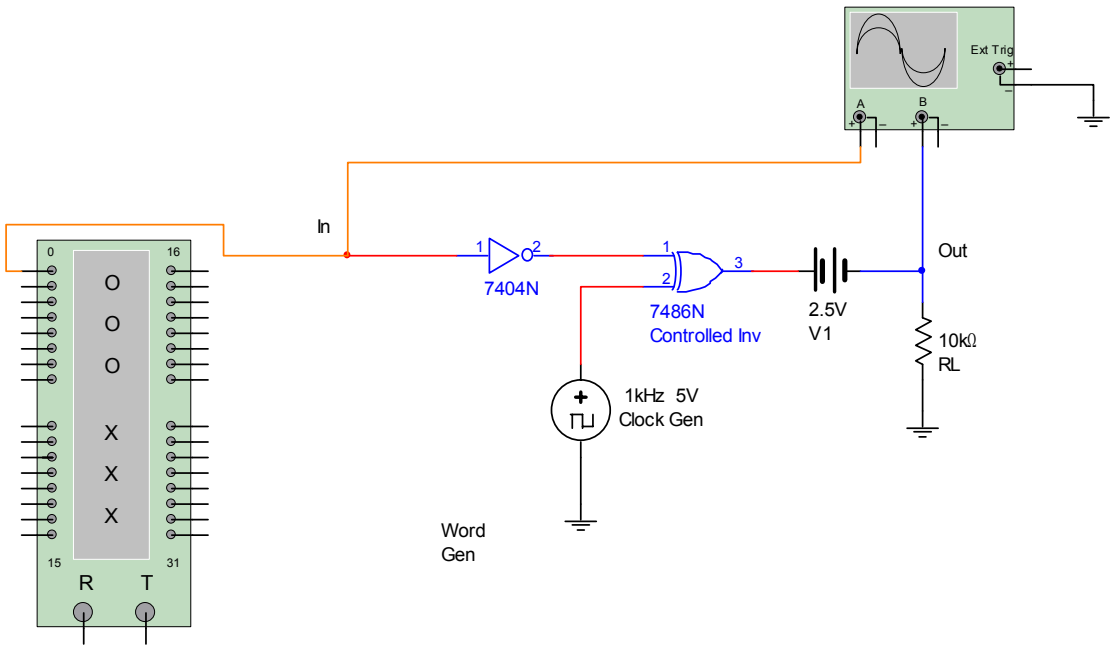
4. Make the next circuit and bring down the oscilloscope enlargement. Make sure that the following settings are selected: Time base (Scale = 1 ms/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = 0, DC), Ch B (Scale = 5 V/Div, Ypos = 0, DC), Trigger (Pos edge, Level = 0, Auto. Double click the word generator to bring down the enlargement and make sure the following settings are selected: Controls (cycle), Display (Hex), Trigger (Internal, Pos edge) and Frequency (1 kHz). Make sure the right hand columns of the first 8 rows of words are respectively entered as 0, 0, 1, 0, 1, 1, 0, 1.Remember to set the final position on the last word. Run the simulation. Make sure that the output curve is symmetrical about the horizontal axis (or adjust the variable potentiometer (R) until it is), and then stop the simulation. Click *burst* to run the simulation again. Draw the 8-bit binary input code and the output curve. Measure the output logic 0 and logic 1 voltage levels. Is the input NRZ or RZ encoded? Is the input unipolar or bipolar? Is the output NRZ or RZ encoded? Is the output unipolar or bipolar? Explain why?



5. Make the circuit below and bring down the oscilloscope enlargement. Make sure that the following oscilloscope settings are selected: Time base (Scale = 1 ms/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = 0, DC), Ch B (Scale = 5 V/Div, Ypos = 0, DC), Trigger (Pos edge, Level = 0, Auto). Double click the word generator to bring the enlargement and make sure the following settings are selected: Controls (Burst), Display (Hex), Trigger (Internal, Pos edge) and Frequency (1 kHz). Make sure the right hand columns of the first 8 rows of words are respectively entered as 1, 0, 0, 1, 0, 0, 1, 1. Draw the 8-bit binary input code and the output curve. Measure the output logic 0 and logic 1 voltage levels. What is the encoder 8-bit binary input code? Is the output NRZ or RZ encoded? Is the output unipolar or bipolar? Explain why?

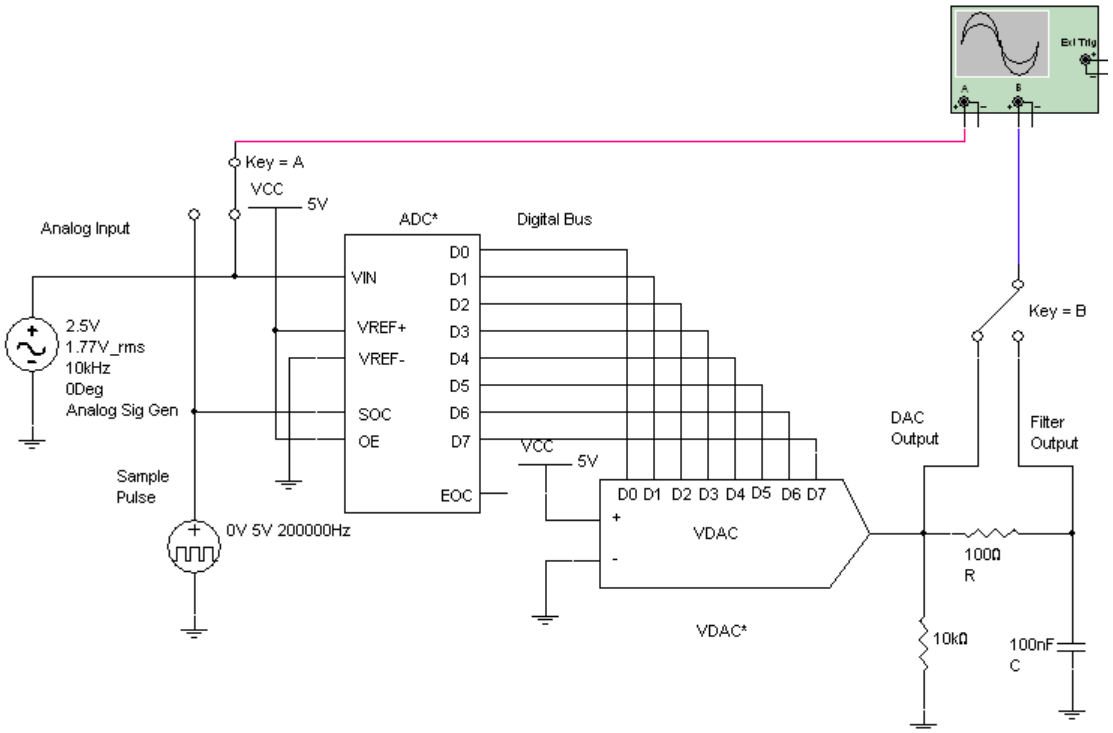


6. Wire the circuit below, on the word generator make sure that the right hand columns of the first 8 rows of words are respectively entered as 1, 0, 0, 0, 1, 0, 1, 1 and repeat step 5. Notice that the encoder output curve has a voltage transition at the centre of each bit interval. What is the encoder 8-bit binary input code? Is the output Manchester (biphase) encoded? Is the output unipolar or bipolar? Explain why?



Part B – PCM

7. Wire the circuit shown below and bring down the oscilloscope enlargement. Make sure that the following settings are selected: Time base (Scale = 20 us/Div, Xpos = 0, Y/T), Ch A (Scale = 2 V/Div, Ypos = 1.0, DC), Ch B (Scale = 2 V/Div, Ypos = -2.0, DC), Trigger (Pos edge, Level = 0, Auto). Run the simulation and draw the displayed analog input signal (red) and the DAC output (blue). Measure the time between samples (Ts) on the DAC output curve plot and calculate the sampling frequency (fs). How did the sampling frequency compare with the frequency of the sampling frequency generator, the analog input frequency and the Nyquist frequency?



8. Click the arrow in the circuit window and press the A key to change Switch A to the sampling generator output. Change the oscilloscope time base scale to 10 us/Div. Run the simulation and draw the plot of the sampling generator output and the DAC output. What is the relationship between the sampling generator output and the DAC staircase output?

9. Change the oscilloscope time base scale back to 20 us/Div. Click the arrow in the circuit window and press the A key to change Switch A back to the analog input. Press the B key to change Switch B to Filter Output. Run the simulation and draw the plot of the analog input and the low-pass filter output displayed on the oscilloscope. What happened to the DAC output after filtering? Is the output wave shape a close representation of the analog input wave shape? Calculate the cut-off frequency (fc) of the low-pass filter and compare it with the analog input frequency.

10. Change the filter capacitor to 20 nF and calculate the new cut-off frequency (fc). Bring down the oscilloscope enlargement and run the simulation to completion again. How did the new filter output compare with the previous filter output? Explain.

DCM 3 – FSK and Parity Checking

OBJECTIVES

1. Demonstrate how digital pulses are converted to an analog signal, and vice versa, using Frequency Shift Keying (FSK).
2. Learn about parity checking as a method of error detection and correction.

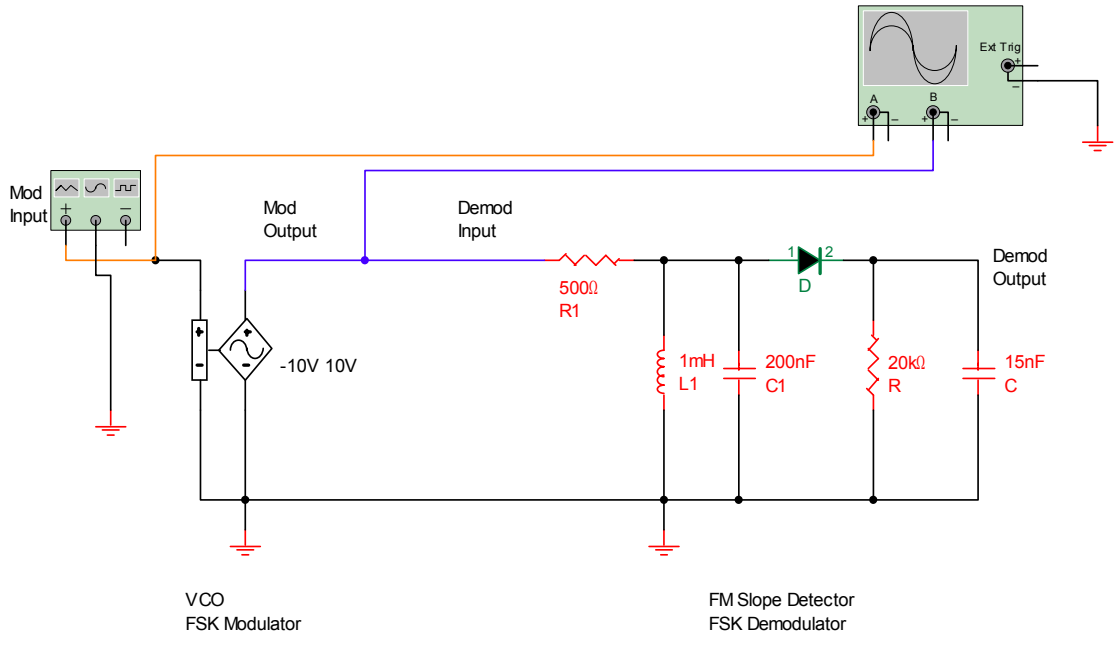
EQUIPMENT

This experiment will use the Electronics Workbench (Multisim 9) simulation.

PROCEDURE

Part A – FSK

1. Wire the circuit shown below and double click the oscilloscope to bring down the enlargement. Make sure that the following settings are selected: Time base (Scale = 500 us/Div, Xpos = 0, Y/T), Ch A (Scale = 5 V/Div, Ypos = -2.0, DC), Ch B (Scale = 10 V/Div, Ypos = 0, DC), Trigger (Pos edge, Level = 0, Auto). Double click the function generator to bring down the enlargement. Make sure the following settings are selected: *Square Wave*, Freq = 250 Hz, Duty Cycle = 50%, Ampl = 2.5 V, Offset = 2.5 V. Double click the voltage controlled oscillator to bring down the enlargement. Make sure the following settings are selected: *Output peak voltage = (*-10V *to* 10V*), control and frequency arrays* = 5V & 0V and 10000Hz & 5000Hz respectively. Run the simulation for one full screen display on the scope and then pause. Draw the FSK modulator input and output as shown on the oscilloscope screen. Is the input a digital or analog wave shape? Is the output a digital or analog wave shape? Explain why.



2. Measure the bit time (t_b) of one of the input bits and calculate the input bit rate (f_b) and baud rate. Assume the input consists of alternating ones and zeros (1010 ---). Expand the oscilloscope horizontal time base by changing the time base scale to 200 us/Div. Measure the modulator output (blue) time period for one cycle when the input is low (T_s) and when the input is high (T_m). Based on these time periods, calculate the modulator output frequencies when the input is low (f_s) and when the input is high (f_m). What is the relationship between the modulator input voltage and output frequency? Based on the input bit rate (f_b), the mark frequency (f_m), and the space frequency (f_s), determine the bandwidth (B) required for the FSK signal.

3. Change the function generator frequency to 500 Hz and run the simulation for one full screen display. Measure the bit time (t_b) of one of the input bits and calculate the input bit rate (f_b) and baud rate. How did the bit rate compare with the bit rate in step 2? Measure the modulator output time period for one cycle when the input is low (T_s) and when the input is high (T_m). Based on these time periods, calculate the modulator output frequencies when the input is low (f_s) and when the input is high (f_m). Did the mark and space frequencies change when the bit rate was changed? Explain why.

4. Move the oscilloscope Channel B input to the demodulator output. Change the function generator frequency back to 250 Hz. Change the oscilloscope horizontal time base scale to 2 ms/Div and Channel B scale to 5V/Div. Run the simulation for one full screen display and draw the modulator input (bottom curve plot) and the demodulator output (top curve plot) as shown on the oscilloscope. Is the modulator input wave shape digital or analog? Is the demodulator output wave shape digital or analog? Was this expected?

5. Change the function generator frequency to 100 Hz and run the simulation for one full screen display. Observe the modulator input and demodulator output. How did the modulator input and the demodulator output change?

6. Move the oscilloscope Channel A input to the demodulator input. Change the function generator frequency back to 250 Hz. Change the oscilloscope horizontal time base scale to 1 ms/Div. Run the simulation for one full screen display and draw the demodulator input (bottom curve plot) and the demodulator output (top curve plot) as shown on the oscilloscope. What is the relationship between the demodulator input frequency and the demodulator output voltage level?

Part B – Parity Checking

7. Wire the circuit below and record the predicted odd and even output logic levels for each input combination in the Table 1 below. Run the simulation. After switching the logic levels to the appropriate positions, record the measured odd and even output logic levels for each input combination.

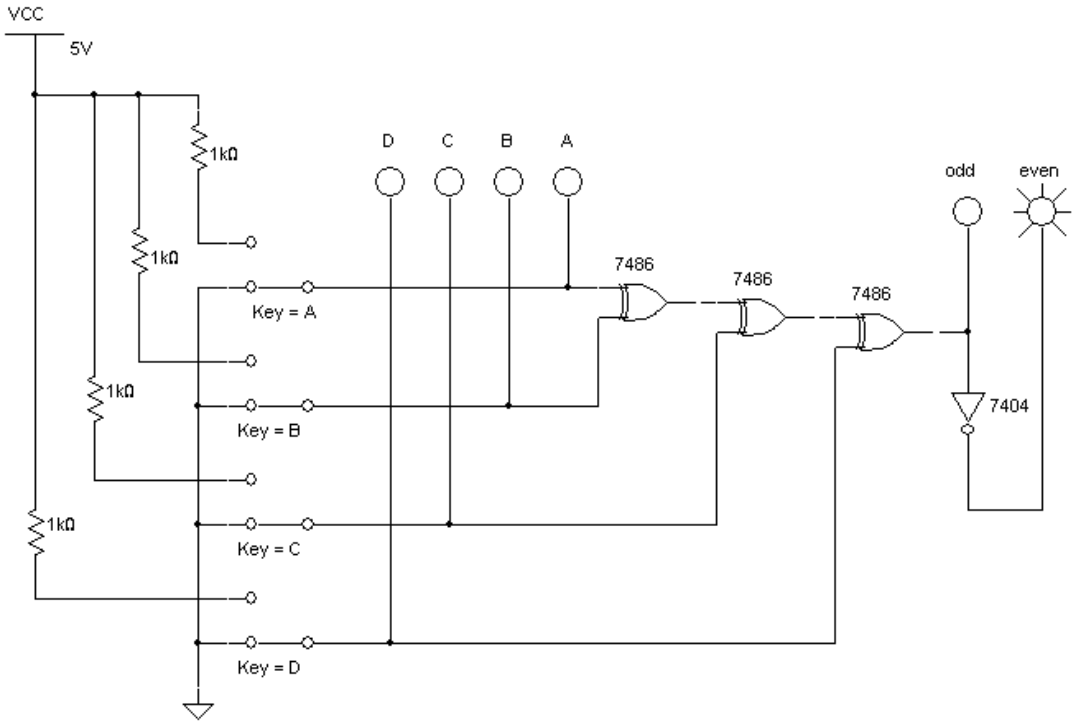

VCC
5V
1kΩ
1kΩ
1kΩ
1kΩ
D
C
B
A
Key = A
Key = B
Key = C
Key = D
7486
7486
7486
odd
even
7404

Table 1				Predicted		Measured	
D	**C**	**B**	**A**	**Odd**	**Even**	**Odd**	**Even**
0	0	0	0				
0	0	0	1				
0	0	1	0				
0	0	1	1				
0	1	0	0				
0	1	0	1				
0	1	1	0				
0	1	1	1				
1	0	0	0				
1	0	0	1				
1	0	1	0				
1	0	1	1				
1	1	0	0				
1	1	0	1				
1	1	1	0				
1	1	1	1				

How did your predicted outputs compare with the measured values? What is the logic circuit in the above circuit doing?

8. Wire the next circuit and record the predicted parity bit output logic levels for each bit combination in Table 2.

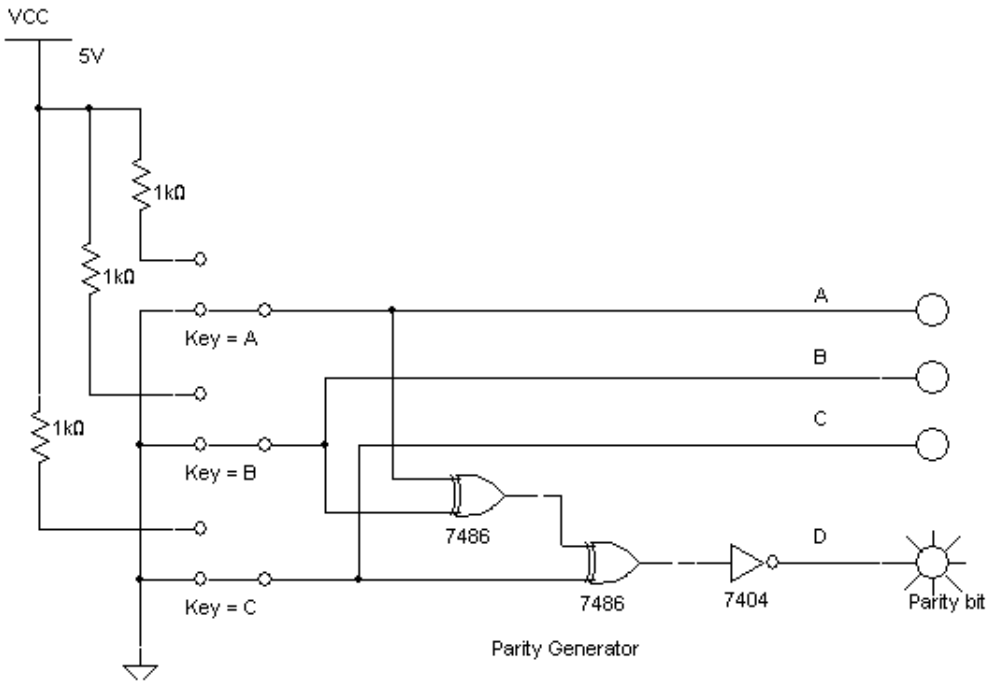
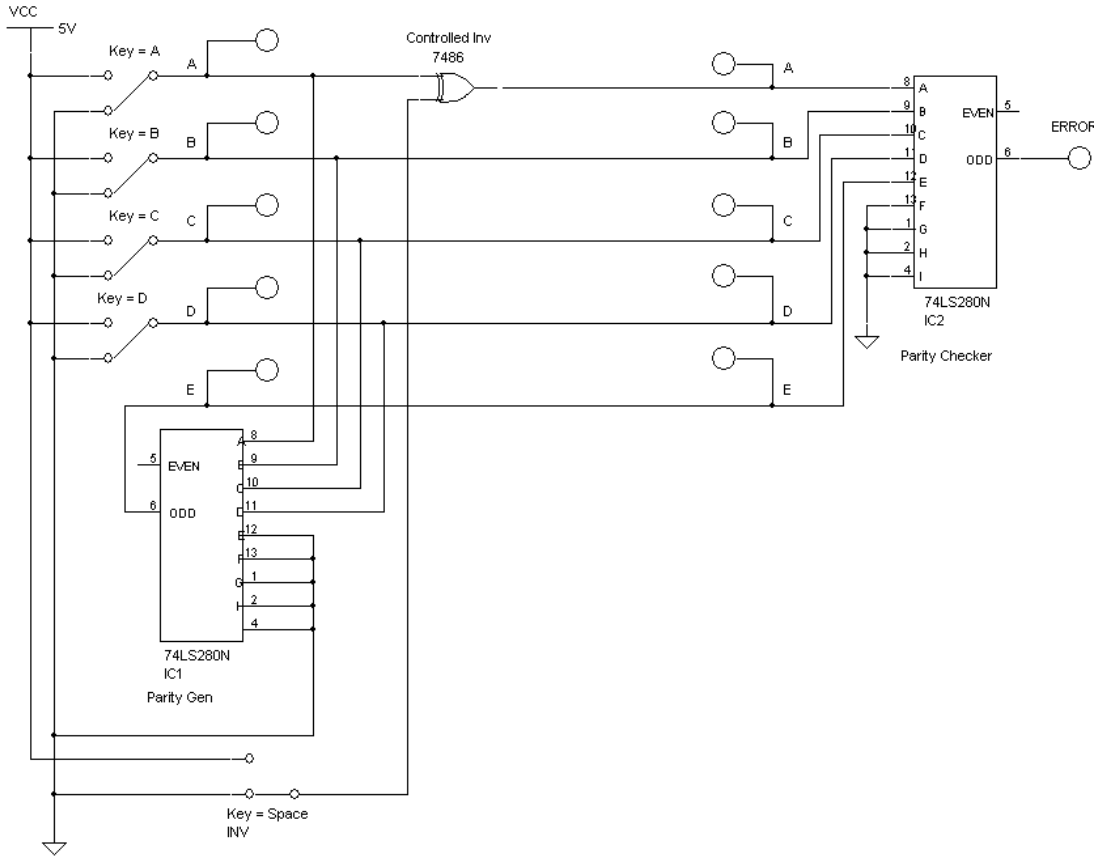


Table 2			Predicted	Measured
C	**B**	**A**	**Parity bit**	**Parity bit**
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Run the simulation. After switching the logic levels to the appropriate positions, record the measured parity bit output logic levels for each input combination. How did your predicted outputs compare with the measured values? Is the logic circuit in above generating an even parity 4-bit code or an odd parity 4-bit code? Explain why.

9. Make the circuit shown below. Notice that a 4-bit binary code will be generated by the settings of switches A, B, C, and D, with a fifth bit (parity bit E) generated by the parity generator (IC1). This 5-bit code will be transmitted over a 5-bit transmission line to a 5-bit parity checker (IC2) on the other end of the line. Also notice that a controlled inverter (7486) has been inserted in one of the transmission lines (line A) to make it possible to simulate a bit error. When the INV switch is down (0), the binary bit on line A is passed through the controlled inverter unchanged. When the INV switch is up (1), the binary it on line A is inverted, causing the output code to have a bit error. Run the simulation and generate various binary codes to test each 4-bit input combination by switching logic switches A, B, C, and D. Were the received binary codes the same as the transmitted binary codes? Did the error light turn on for any of the transmitted codes?



10. Press the SPACE BAR on the keyboard to switch the INV to the up (1) position. Generate various binary codes to test each 4-bit input combination by switching logic switches A, B, C, and D. Were the received binary codes the same as the transmitted binary codes? Did the error light turn on for any of the transmitted codes? Is this an even or odd parity error-detection system? Explain the reason for your answer.

F. Semiconductor Electronics – EE3311

SCE1 – Characteristics of a Thermistor

INTRODUCTION

This experiment is aimed to give an idea about the band gap measurements of a semiconductor and also to study the characteristics of a thermistor. Thermistor or thermally dependent resistors are devices, which make use for the large temperature coefficient of resistivity of a semiconductor. They are normally made from semiconduting oxides with controlled amounts of impurities. Two common types of thermistors are:-

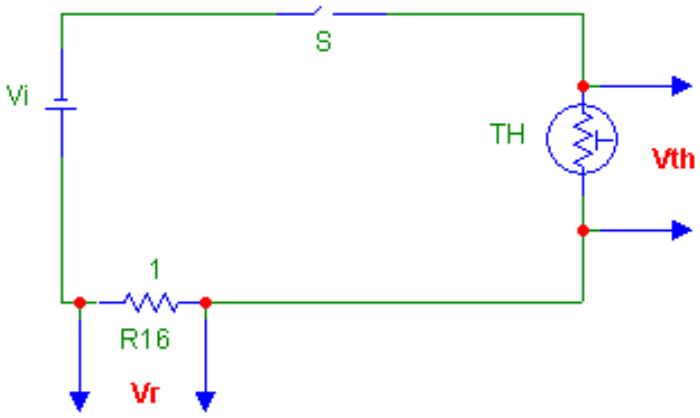
i) Directly heated thermistors and
ii) Indirectly heated thermistors.

In a directly heated thermistor, as voltage is applied to the terminal of the thermistor, a current will flow which is voltage divided by the room temperature resistance of the thermistor. Power will be dissipated in the thermistor in the amount VI. But since V and I are small, power will have only negligible heating effect. As V is increased however, the power gets larger and the thermistor begins to heat up. But as the temperature increases resistivity decrease and for a given voltage, current should be large. Thus uses of the thermistor device pertain to the large variation of resistance with temperature and its Voltage-Current characteristics.

In an indirectly heated thermistor, the semiconductor element is usually a bead surrounded by a tiny heating coil. The bead and the coil are held in proper position in a glass capsule to produce a four terminal device.

EXPERIMENT

The experimental procedure is with reference to the figure below and the fill-in the table: -



1. Measure V_R and V_{TH} at Room Temperature i.e. before turning the heater ON. Hence calculate R_{TH} at the room temperature.

2. Take readings corresponding to Temperature at 10°C interval), V_R and V_{TH}.
3. Calculate corresponding R_{TH} values for 2. Above.
4. Determine the I-V and ρ-T characteristics of the thermistor.
5. Calculate the resistance of the thermistor at 60°C.
6. Discuss the difference in the "theoretical" vs "experimental" curves obtained.
7. Calculate the ***band-gap*** (or energy-gap) of the thermistor.

TEMP.	VOLTAGE ($\mathbf{V_R}$)	VOLTAGE ($\mathbf{V_{TH}}$)	Calculated Value of The Thermistor ($\mathbf{R_{TH}}$)